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Abstract

Cyber-deception is a proactive defense paradigm that seeks to rebalance the asymmetry between attackers and defenders. Two decades of research have produced a rich but fragmented body of surveys with incompatible taxonomies, heterogeneous scopes, and uneven methodological rigor. We present the first systematic meta-survey of cyber-deception surveys to provide a unified view of the field. We group existing work into three families: general cyber-deception surveys, moving target defense surveys, and honeypot surveys. Our analysis shows that current classification models are siloed within these families and lack a shared vocabulary that spans the full spectrum of deception technologies. We also find that important techniques such as tarpits, honey encryption, and honey patches are largely absent from existing systematic surveys and their taxonomies. To address these gaps, we introduce a comprehensive taxonomy grounded in the MITRE Engage framework that integrates disparate models into a single cohesive structure. We further synthesize future research directions, highlighting the growing role of AI and machine learning in deception. Finally, we identify a persistent foundational gap: the absence of standardized evaluation frameworks and quantitative metrics for effectiveness, performance overhead, and cost benefit trade offs, which undermines claims of superiority, deployment decisions, and assessment of community progress.

A Systematic Meta-Survey of Cyber Deception: Unified Taxonomy and Research Directions

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CCS Concepts: • **Security and privacy** → **Network security**.

Additional Key Words and Phrases: cyber-deception, honeypot, deception, survey, meta survey, moving target defense

1 INTRODUCTION

The modern digital landscape is defined by a persistent and asymmetric conflict between attackers and defenders. For decades, the foundational paradigm of cybersecurity has relied on a *castle-and-moat* model centered on static defenses such as firewalls, intrusion detection systems, and antivirus software. However, a series of high-impact cyber attacks has systematically demonstrated that this defensive model is fundamentally inadequate against modern threats: the MIRAI botnet weaponized Internet of Things (IoT) devices for massive distributed denial-of-service attacks [5], large-scale ransomware outbreaks such as WannaCry caused widespread disruption, the SolarWinds (Sunburst) campaign subverted trusted software updates, the TRITON malware targeted industrial safety systems, and the Log4Shell vulnerability exposed a near-universal attack surface [62].

These incidents collectively reveal a fundamental truth about modern cybersecurity: Static prevention-centric defenses are necessary but insufficient. The security community faces an inherent asymmetry. Defenders must successfully protect every potential entry point and asset, while attackers need only identify and exploit a single weakness [13, 72, 75]. Consequently, the field has increasingly adopted an *assume breach* mindset that acknowledges perimeter penetration as inevitable [40, 55]. In response, cybersecurity research has shifted toward dynamic and proactive defense strategies designed to reverse this asymmetry. Cyber-deception has emerged as a promising paradigm that moves beyond constructing impenetrable barriers to actively misleading, confusing, and entrapping adversaries who have already penetrated the network perimeter [37]. Drawing on centuries of military deception doctrine, cyber-deception involves

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the deliberate introduction of false, misleading, or ambiguous information and resources to manipulate attacker perceptions, decisions, and actions [13, 37].

The cyber-deception landscape spans a diverse array of techniques serving distinct strategic objectives. Honey tokens [8] and other fake credentials, database entries, or API keys act as high-fidelity tripwires for detection; interactive honeypots [60] and honeynets [61] provide controlled environments to observe and analyze adversary behavior [72]; tarpits deliberately slow connection attempts to exhaust automated scanning tools and botnets [3]; and Moving Target Defense (MTD) strategies dynamically alter the attack surface, invalidating reconnaissance data and forcing adversaries to re-scan and re-target systems [13, 55]. Beyond these technical mechanisms, advanced deception strategies exploit human cognitive biases through breadcrumbing [47], signaling [18], and psychological manipulation to guide attacker behavior toward defensive objectives [19, 58]. We provide a detailed overview and systematization of these in Sec. 3.

Over the past two decades, both academic research and practical deployment of cyber-deception techniques have grown substantially, generating an extensive body of primary research literature. This expansion has, in turn, spawned a significant secondary wave of survey papers, systematic reviews, and mapping studies attempting to organize and synthesize the field’s knowledge. However, this survey literature itself has become increasingly fragmented and difficult to navigate. Researchers and practitioners entering the field now confront dozens of surveys, each employing different scopes, incompatible classification schemes, and varying levels of methodological rigor.

This fragmentation creates substantial obstacles to developing comprehensive understanding. Certain subfields, particularly MTD and honeypots, have received extensive survey coverage, with multiple papers examining overlapping topics from different perspectives. Conversely, important deception techniques including tarpits, honey encryption, and bias exploitation strategies remain largely absent from systematic analysis. Different surveys propose incompatible taxonomic structures, making it difficult to compare findings or integrate insights across the literature. Many surveys focus primarily on enabling methodologies such as game theory or machine learning applications rather than comprehensively examining deception techniques themselves. Furthermore, methodological quality varies dramatically, with numerous surveys lacking explicit literature search strategies, inclusion criteria, or quality assessment procedures, limitations that compromise their reliability and reproducibility.

To our knowledge, no prior work has provided a systematic meta-analysis to synthesize this fragmented survey landscape and develop a holistic understanding of how cyber-deception research has been organized, what gaps persist, and where the field should direct future efforts. Our goal is not to resurvey all primary deception research, but to critically examine and unify the existing survey layer itself, which has become a bottleneck for understanding the field due to incompatible taxonomies and weak methodologies. This paper addresses that gap by presenting the first comprehensive meta-survey of the cyber-deception survey literature, systematically analyzing how the field has surveyed, organized, and synthesized itself. We employ a rigorous methodology based on the PRISMA framework [64] to identify 258 candidate papers, screen them, and conduct an in-depth analysis of 46 survey papers spanning general cyber-deception, MTD-specific topics, honeypot research, and related areas. Our analysis reveals the field’s evolution, assesses taxonomic approaches, identifies coverage gaps, evaluates methodological rigor, and synthesizes persistent research challenges. Our contributions are the following:

- **Systematic Taxonomy Analysis:** We provide the first comprehensive comparative analysis of cyber-deception taxonomies, examining classification frameworks proposed for general cyber-deception, MTD, and honeypots. This analysis identifies common themes, fundamental divergences, and opportunities for integration (Sec. 4).

- **Comprehensive Meta-Analysis of Survey Literature:** We conduct an in-depth meta-survey of general cyber-deception (Sec. 5.1), MTD (Sec. 5.2), and honeypot surveys (Sec. 5.3), analyzing research themes, temporal trends, methodological rigor, and key challenges in each area. We identify significant gaps where important techniques such as tarpits, honey encryption, and honey patches receive little or no survey coverage, and critically assess survey methodology (literature search strategies, inclusion criteria, quality assessment procedures), exposing widespread weaknesses that limit reliability and reproducibility. Finally, we synthesize open challenges and future research directions, highlighting consensus on standardized evaluation frameworks, comprehensive metrics, AI integration, hybrid system development, real-world validation, and human factors research (Sec. 5).
- **New Unified Taxonomy:** Building on our comprehensive analysis of existing taxonomies and identified gaps, we propose a new, unified taxonomy for cyber-deception that integrates disparate models from the literature into a single, comprehensive framework addressing the limitations of prior classification schemes (Sec. 6). We also provide an interactive online version of this taxonomy to help researchers and deception operators explore technique options and make better design choices¹.

Taken together, these contributions yield a unified taxonomy that is not merely a relabeling of existing schemes but novel in three ways. First, it consolidates taxonomic schemes from general cyber-deception, MTD, and honeypot research into a single cross-domain framework. Second, it elevates psychological and cognitively oriented deception, together with underexplored techniques such as bias exploitation, tarpits, honey encryption, honey patches, and honeytokens, to first-class elements of the design space. Third, it explicitly aligns deception goals and deployment locations with the MITRE Engage [17] and MITRE ATT&CK [16] frameworks, enabling integration into operational workflows and security tooling.

2 METHODOLOGY

Generally, literature reviews come in two forms: *traditional* and *systematic*. Traditional methods are mostly subjective, relying on the author’s expertise to select the most representative work in the field to present a topic. These methods are useful to describe particular issues and concepts from a narrow perspective, but are difficult to reproduce and often suffer from bias and systematic errors [42]. Systematic literature reviews provide formal and structured approaches to select, review, and extract concepts from the literature, reducing the risk of bias and increasing the reproducibility of the study [46]. Nowadays, there are many established methods to conduct systematic literature reviews for other areas of science, such as PRISMA [42, 46] and SALSA [29].

From the available frameworks, PRISMA is one of the most widely used and rigorously validated reporting standards for evidence-synthesis studies: It provides a transparent, reproducible workflow for identification, screening, eligibility, and inclusion, and has been successfully adapted beyond healthcare into computer science, and software engineering [41, 53, 64]. Therefore, we adopted PRISMA as the methodological basis for our review.

2.1 Eligibility criteria

We used Scopus [56] as our primary bibliographic indexer to identify candidate surveys on cyber-deception. The search targeted publication types explicitly labeled as surveys, taxonomies, reviews, or SoKs, and accepted linguistic variants of core deception-related terms. Matching terms included decoy, canary, honey*, deception, deceptive, MTD, breadcrumbs, and tarpit, together with modifiers and expanded forms (e.g., honeypots, honey tokens, moving

¹Our interactive taxonomy is available at <https://deception.compute.dtu.dk/>.

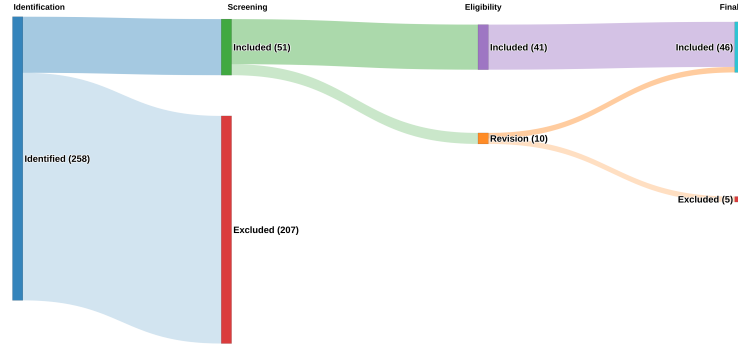


Fig. 1. PRISMA Flow Diagram for study Identification, Screening, Eligibility, and Inclusion.

target {defense OR defence}). Only studies published before 2025 were considered. Applying this query yielded 258 records (Identification). The exact Scopus query is provided in the Appendix (Listing 1).

Two reviewers (R1 and R2) independently screened all records based on titles, abstracts, and, when necessary, a brief inspection of the full content. Reviewers applied a lightweight quality-assessment protocol informed by PRISMA, the Critical Appraisal Skills Programme (CASP) [20]—a set of widely used checklists for assessing the rigor, credibility, and relevance of empirical studies—and established methodological guidance [11, 39, 41, 53]. The full set of appraisal questions is listed in the Appendix (Table 8). Papers mislabeled as surveys/taxonomies/SoKs or lacking a section on defensive deception were excluded. In addition, reviewers removed 13 papers for language reasons (12 written in Chinese and 1 in Turkish). After this initial screening, 51 papers remained. From these, R1 and R2 agreed on 41 papers. The remaining 10 conflicting items (3 exclusive to R1, 7 exclusive to R2) were forwarded to a third reviewer (R3). Reviewer R3 resolved the classification conflicts by excluding five papers from the list, yielding 46 studies for the final qualitative synthesis. These were grouped into three conceptual categories commonly represented across the literature: cyber-deception, honeypots, and Moving Target Defense (MTD). No additional searches (e.g., backward/forward snowballing) were performed beyond this stage. A summary of the identification, screening, eligibility, and inclusion steps is shown in Fig. 1. Restricting the search to Scopus and English-language surveys offers a reproducible baseline and captures the dominant body of work, but may omit some regional or venue-specific surveys; we therefore interpret our findings as characterizing the main trends in the indexed literature rather than claiming exhaustive global coverage.

2.2 Review process

The review team consisted of six reviewers. The remaining papers were then shuffled and allocated using pairwise permutations of reviewers, guided by maximum Euclidean distance to evenly distribute the workload. This strategy helped mitigate potential biases, as rigid reviewer groups may otherwise form opinionated islands, whereas shuffling promotes diverse perspectives through varied reviewer pairings.

Each reviewer independently examined their assigned studies to determine the paper type, topic, key contributions, and the research, analysis, and evaluation methods employed. All surveys were additionally annotated using a shared code book, designed to capture common characteristics, support the development of a unified deception taxonomy, and ensure consistent assessment across reviewers. The code book evolved during the process as reviewers encountered new

Group	Label	Code	Description
Taxonomy	Taxonomy present	TY1	Study includes or proposes a taxonomy or classification
Topic	Cyber-deception	TP1	Focuses on cyber operations, traps, or deceptive cyber systems
	Honeypot	TP2	Focuses on honeypot mechanisms
	MTD	TP3	Concerns Moving Target Defense mechanisms
Research method	Quantitative	RM1	Empirical, measurement-based, or statistical analysis
	Qualitative	RM2	Conceptual analysis, narrative synthesis, or descriptive study
Review method	Traditional	RE1	Narrative review without systematic protocol
	Systematic	RE2	Review following explicit methodology
Evaluation	Evaluation present	EV1	Study includes evaluation or comparison

Table 1. Codebook used to annotate and classify defensive-deception surveys.

characteristics and was subsequently harmonized before merging the individual assessments. The final consolidated version is presented in Table 1.

Although our own review procedure incorporates structured assessment, multi-reviewer triangulation, and a code book, such a methodological approach is mostly absent in prior cyber-deception surveys. As detailed in Table 2, existing surveys rarely document their screening strategies or quality-assessment processes, and when methodology is provided, it often relies on simplistic keyword searches in single indexers [67] or opaque, non-reproducible reviewer decisions [22]. These weaknesses contribute to inconsistent coverage and, as noted by Han [31], can result in biased or incomplete representations of the available evidence. This broader context motivated the design of our multi-reviewer workflow and unified coding protocol.

After individual assessments were completed, reviewers jointly merged their evaluations using the annotations and the standardized code book. Table 2 provides the complete list of the reviewed surveys, together with their final classification and identified characteristics. In addition to merging reviewers' assessments, we included the publication year of each study to observe how methodological practices have evolved over time. As shown in Table 2, the chronological distribution reveals that, despite an increasing number of surveys published in recent years, there is no corresponding growth in methodological rigor: Only a small fraction employ systematic review procedures, and even fewer incorporate any form of empirical evaluation. Several of the most recent surveys (2021–2024) continue to rely exclusively on narrative or descriptive methods, with little evidence of quantitative analysis, structured appraisal, or reproducible methodology. This absence of methodological progression underscores the need for more rigorous and empirically grounded survey practices in the cyber-deception domain.

3 BACKGROUND: CONCEPTS AND TECHNIQUES

Cyber-deception is a proactive defense paradigm that complements traditional reactive, perimeter-based mechanisms such as firewalls and anti-virus software [72, 75]. Rather than focusing solely on blocking or detecting attacks, cyber-deception deliberately introduces ambiguous, false, or misleading information and resources into a system to manipulate

Study	Tax.	Topic			Research Method		Review Method			Year
		CD	HP	MTD	Qual.	Quant.	Trad.	Syst.	Eval	
Zhang et al. [72]	✓		✓		✓			✓		2003
McQueen and Boyer [45]	✓	✓			✓		✓			2009
Al-Gharabally et al. [2]			✓		✓		✓			2009
Ge et al. [26]				✓	✓			✓		2014
Heckman and Stech [33]	✓	✓			✓		✓		✓	2015
Fang et al. [23]	✓			✓	✓		✓			2015
Farris and Cybenko [24]				✓	✓		✓			2015
Briskin et al. [9]	✓	✓			✓					2016
Cai et al. [10]	✓			✓		✓	✓		✓	2016
Hassan and Guha [32]		✓			✓			✓		2017
Urias et al. [68]	✓	✓			✓		✓		✓	2017
Faveri et al. [25]		✓				✓		✓	✓	2017
Khosravi-Farmad et al. [40]	✓			✓	✓		✓			2018
Lei et al. [44]	✓			✓	✓		✓			2018
Zheng and Namin [75]				✓	✓		✓			2019
Han et al. [31]		✓			✓			✓	✓	2019
Chen et al. [12]				✓		✓	✓			2019
Efendi et al. [22]		✓				✓	✓			2019
Pawlick et al. [52]	✓	✓				✓				2020
Sengupta et al. [57]	✓			✓		✓	✓		✓	2020
Gonzalez et al. [28]		✓			✓		✓		✓	2020
Torquato and Vieira [67]				✓	✓		✓		✓	2020
Cho et al. [13]	✓			✓	✓		✓			2020
Navas et al. [50]				✓		✓		✓	✓	2021
Zhang et al. [73]	✓			✓	✓		✓			2021
Zhang and Thing [74]		✓			✓			✓		2021
Zhu et al. [77]	✓	✓			✓		✓			2021
Jalowski et al. [36]	✓			✓	✓			✓		2022
Zheng et al. [76]	✓			✓	✓					2022
Mohan et al. [48]	✓	✓			✓		✓			2022
Nguyen and Debroy [51]	✓			✓		✓	✓			2022
Saputro [55]				✓	✓		✓			2023
Amro et al. [4]	✓			✓	✓		✓			2023
Sun et al. [65]	✓			✓	✓		✓			2023
Tan et al. [66]				✓		✓	✓			2023
Deshpande and Gupta [21]		✓			✓			✓	✓	2023
Qin et al. [54]		✓				✓		✓	✓	2023
Antunes and Sanchez [6]		✓			✓		✓			2023
Yi et al. [71]		✓			✓		✓			2023
Silaen et al. [59]			✓		✓			✓		2023
Javadpour et al. [37]	✓		✓		✓		✓		✓	2024
Chouhan and Aujla [14]	✓	✓				✓	✓			2024
Lee et al. [43]				✓	✓		✓			2024
Comney et al. [15]	✓	✓			✓		✓		✓	2024
Abdi et al. [1]				✓	✓		✓			2024
Haighton and Leblanc [30]		✓			✓		✓			2024

Table 2. Studies sorted by publication year (ascending). A checkmark (✓) indicates that the corresponding feature is present. Topic columns: CD = Cyber-Deception, HP = Honeypot, MTD = Moving Target Defense. Review Method: Traditional, Systematic.

an attacker's perceptions and actions [13, 37]. By exploiting the inherent information asymmetry in cybersecurity, where defenders know their own systems while attackers must infer them through reconnaissance [40], cyber-deception seeks to mislead, confuse, and entrap adversaries who have already breached the perimeter [13, 55, 75]. Intentionally providing deceptive artifacts enables defenders to increase attacker uncertainty, guide adversaries toward predetermined paths, and gain critical intelligence about attack methodologies.

3.1 Goals of Cyber-Deception

Cyber-deception techniques serve multiple strategic objectives that collectively strengthen an organization's security posture.

3.1.1 Improving Detection of Malicious Activities. One of the most immediate benefits of cyber-deception is enhanced threat detection. Traditional security systems often struggle with distinguishing malicious activity from legitimate user behavior, leading to alert fatigue among security teams. Deception-based detection mechanisms address this challenge by creating resources that have no legitimate use case: Any interaction with these resources is, by design, suspicious or malicious [59, 72].

This approach provides high-fidelity alerts that warrant immediate investigation. Unlike traditional anomaly detection systems that must establish baselines of normal behavior and risk flagging legitimate but unusual activities, deception-based detection offers binary clarity: Authorized users have no reason to access deceptive resources, so any access represents a security event. This dramatically improves the signal-to-noise ratio in security operations centers, allowing analysts to focus their limited time and attention on genuine threats rather than sorting through thousands of potential false positives [8, 37].

3.1.2 Intelligence Gathering on Attacker Capabilities and Intentions. Beyond detection, cyber-deception provides a unique opportunity to observe attacker behavior in a controlled environment, gathering high-value threat intelligence that informs defensive strategies [2, 72]. When attackers interact with deceptive resources, defenders can monitor their tactics, techniques, and procedures (TTPs) in real-time without risking actual assets or alerting the adversary to their observation.

This intelligence serves multiple purposes. First, it reveals the skill levels and sophistication of the attacker, helping organizations understand the level of risk. Second, it exposes the specific tools, exploits, and methodologies employed by adversaries, enabling defenders to patch vulnerabilities, update detection signatures, and strengthen defenses against observed attack patterns. Third, it provides insights into attacker objectives and motivations by revealing which systems they target, what data they seek, and how persistent they are in pursuit of their goals. This strategic intelligence is invaluable for threat modeling, risk assessment, and resource allocation decisions [37, 59].

3.1.3 Wasting Attacker Time and Resources. Cyber-deception imposes costs on adversaries by consuming their limited time, computational resources, and attention. In the economic model of cybersecurity, defenders seek to increase the cost of successful attacks to the point where they become economically nonviable or operationally impractical [13, 27]. Deception achieves this by introducing friction into the attack process.

When attackers encounter deceptive resources, they must invest effort in exploring them, attempting exploitation, and ultimately determining whether they represent genuine targets or decoys. This time spent on false leads is time not spent advancing toward real objectives. For automated attacks and botnets that rely on speed and scale, even small delays per target can significantly reduce overall campaign effectiveness.

3.1.4 Increasing Attacker Uncertainty and Operational Risk. Perhaps the most strategic benefit of cyber-deception is its ability to fundamentally undermine attacker confidence and decision-making. When adversaries cannot reliably distinguish between genuine and deceptive resources, their reconnaissance data becomes suspect, their situational awareness degrades, and every action carries the risk of revealing their presence to defenders [55, 75].

This uncertainty compounds throughout the attack lifecycle. Attackers must question whether discovered vulnerabilities are real or honeypots, whether stolen credentials will provide access or trigger alerts, whether exfiltrated data is genuine or fabricated, and whether observed system configurations reflect reality or deliberate misdirection. This cognitive burden slows attack progression, increases the likelihood of operational mistakes, and forces adversaries to adopt more cautious (and therefore likely more time-consuming and detectable) approaches [44, 66].

3.2 Cyber-Deception Techniques

Different cyber-deception techniques excel at achieving specific objectives, though many provide multiple benefits simultaneously.

3.2.1 Honeypots and Honeynets. Honeypots are functional, but decoy systems intentionally configured with apparent vulnerabilities to attract and engage attackers [60]. They range from low-interaction honeypots that simulate limited services (such as SSH or HTTP servers responding to basic commands) to high-interaction honeypots that provide complete operating systems and applications for attackers to explore. Honeynets extend this concept by creating entire networks of interconnected honeypot systems that simulate realistic organizational infrastructure [61].

The primary value of honeypots lies in their intelligence-gathering capabilities. By providing a safe, sandboxed environment where attackers can operate freely while being comprehensively monitored, defenders gain unparalleled visibility into adversary behavior. Every command executed, file accessed, tool uploaded, and lateral movement attempt is logged for analysis. This produces rich datasets about attacker TTPs that inform threat models, guide security investments, and enable proactive defense improvements. The effectiveness of honeypots depends critically on their realism [63, 69] and proper isolation. They must be convincing enough that attackers invest significant effort in exploitation, yet sufficiently isolated that compromised honeypots cannot be used as pivots to attack genuine systems.

3.2.2 Honey Tokens. Honey tokens are pieces of false but believable information strategically placed within systems to act as high-fidelity tripwires, with the primary goal of improving detection [8]. These can include fake database entries, API keys, internal URLs, or user credentials. The key characteristic of honey tokens is that they are monitored and the interaction with them triggers an alarm. Another important aspect is that they are designed to be attractive to attackers who have gained some level of access but are indistinguishable from legitimate data through automated scanning or passive observation.

Honey tokens are particularly effective for detection because they require no complex behavioral analysis: Any access or use of a honey token is definitively malicious, providing immediate, actionable alerts with virtually zero false positives.

3.2.3 Tarpits. Tarpits are services deliberately designed to respond extremely slowly to connection attempts, effectively trapping automated tools and consuming attacker computational resources [3]. Network tarpits operate at various protocol layers. An SSH tarpit might accept connections but then delay authentication handshakes indefinitely, tying up scanning tools. Web server tarpits might serve content one byte at a time or introduce long delays between HTTP

responses. In each case, the objective is to make attacking systems appear available while actually imposing severe time and resource penalties on attackers who attempt interaction.

Tarpits are effective against automated attacks that rely on speed and parallelism. A vulnerability scanner expecting rapid responses may allocate threads or processes to each target; tarpits can exhaust these resource pools by refusing to complete connections quickly.

3.2.4 Moving Target Defense (MTD). Moving Target Defense represents a fundamentally different approach to cyber-deception that focuses on making the attack surface unpredictable rather than presenting false targets [35]. MTD techniques dynamically and continuously alter system properties such that attackers' reconnaissance data rapidly becomes obsolete, forcing them to repeatedly re-scan and re-target systems while increasing their risk of detection. MTD operates across multiple dimensions. Network-layer MTD techniques randomize IP addresses, shuffle port numbers, or dynamically remap network topologies, ensuring that network scans quickly lose validity. Host-layer MTD approaches rotate operating system configurations, randomize memory layouts, or migrate virtual machines between physical hosts. Application-layer MTD techniques diversify software implementations, randomize execution paths, or mutate code structures to prevent exploit reuse. Data-layer MTD methods encrypt or transform data formats to prevent unauthorized interpretation. The temporal dimension of MTD is equally important. Proactive MTD changes configurations on predetermined schedules, ensuring regular invalidation of attacker reconnaissance regardless of detected threats. Reactive MTD responds to detected anomalies or attacks by immediately changing configurations to disrupt ongoing operations and force attackers to restart their campaigns.

MTD's effectiveness stems from its impact on the attack economics and the cyber kill chain [34]. Attackers typically invest significant effort in reconnaissance to map networks, identify vulnerabilities, and plan exploitation strategies. MTD undermines this investment by ensuring reconnaissance data has a limited shelf life. This forces attackers to choose between continuously re-scanning (increasing their detection risk and resource expenditure) or acting on potentially stale information (increasing their failure risk).

3.2.5 Honey Patches. Unlike traditional patching practices that aim to eliminate all vulnerabilities as quickly as possible, honey patching strategically manages vulnerability disclosure and remediation to create controlled deception opportunities [7]. Its main goal is to frustrate an attacker's ability to definitively determine whether their exploit attempt has actually succeeded or failed on the true system. The core deception occurs when an attempted exploit is detected. Instead of blocking the attack outright, the honey patch executes a transparent and highly efficient redirection, funneling the attacker's efforts toward an unpatched decoy system. This crucial step allows the exploit to appear successful to the attacker, maintaining the illusion that the main system has been compromised while keeping it perfectly safe.

Once the attacker is successfully engaged with the decoy, it begins to serve as a high-value intelligence asset. This decoy is equipped with aggressive software monitors that are dedicated to collecting critical information about the attacker's tools, techniques, and movements.

3.2.6 Honey Encryption and Data Manipulation. A more data-centric approach to deception involves honey encryption and related data manipulation techniques. These methods protect sensitive information by ensuring that attackers who successfully steal encrypted data cannot determine whether they have obtained genuine content or decoys. Unlike traditional encryption that returns random-looking data when decrypted with incorrect keys, honey encryption produces plausible-looking but false plaintext for every possible decryption key, making it impossible for attackers to verify successful decryption [38].

This approach is particularly valuable for protecting high-value data such as password databases or intellectual property. When attackers attempt brute-force decryption, every tried key produces believable results, preventing them from knowing when, or if, they have succeeded. Frameworks like [49] align with this strategy, leveraging deception to create uncertainty and undermine the value of exfiltrated data if analyzed with automatic tools such as document clustering and topic modeling, thus protecting them even after they have been exfiltrated.

3.2.7 Bias Exploitation. Beyond technical deception mechanisms, sophisticated defensive strategies exploit human cognitive biases and decision-making patterns to manipulate attacker behavior [19, 58]. This psychological dimension of cyber-deception recognizes that attackers, particularly human operators, make decisions under uncertainty and are subject to the same cognitive limitations and biases that affect all human judgment. A crucial aspect of bias exploitation is that it operates on two levels: disinformation (deceptive signals) and factual information (truthful but strategically revealed signals). Disinformation involves actively projecting false information to mislead attackers about system characteristics, defensive capabilities, or organizational posture. Conversely, selectively revealing truthful information can also serve defensive deception objectives by manipulating attacker decision-making through carefully controlled transparency. This approach exploits the fact that attackers must constantly assess risk, allocate resources, and prioritize targets, decisions that can be influenced by specific revelation of genuine facts. The strategic use of factual signaling relies on information asymmetry and selective emphasis. While the information conveyed is true, defenders control which truths are visible, their context, and their prominence. These psychological concepts are operationalized by two primary mechanisms: *breadcrumbing* and *signaling*.

- **Breadcrumbing.** Breadcrumbing techniques place strategically crafted false clues, hints, or apparent vulnerabilities that lead attackers along predetermined paths toward defensive objectives [47]. Unlike passive deception mechanisms that merely present false targets, breadcrumbing actively shapes attacker decision making by exploiting natural information seeking behavior. False vulnerability indicators can be planted to draw attention toward honeypots or away from high value assets. Breadcrumbing can involve any type of data, often combining multiple deception techniques as part of a coordinated trail.
- **Signaling.** Signaling techniques involve deliberately communicating information to attackers, to influence their beliefs, decisions, and actions [18]. Unlike breadcrumbs that attackers discover through exploration, signals are actively transmitted to shape attacker perception and behavior. This can be dynamic changes during the attack or injected notifications.

4 TAXONOMY REVIEW

A critical step toward developing a comprehensive understanding of cyber-deception is analyzing how it has been classified and organized in the literature. Over the years, numerous taxonomies have been proposed to structure the vast landscape of deceptive techniques. These approaches vary in scope: some address cyber-deception at a high conceptual level, while others focus on specific subfields, namely MTD and Honeypots. The prominence of these two areas is reflected in a substantial body of literature that includes entire surveys dedicated exclusively to them.

To provide a structured overview, this section first examines taxonomies that address cyber-deception in general terms, then explores the more specialized frameworks developed for MTD and honeypots.

4.1 General Cyber-Deception

The cyber-deception literature presents multiple taxonomies, each offering distinct perspectives for classifying and understanding deceptive techniques. Analysis of existing works reveals that these taxonomies can be organized into five principal categories: (1) taxonomies derived from military doctrine and strategic theory, (2) taxonomies employing multidimensional classification frameworks, (3) taxonomies inspired by biological systems and evolutionary principles, (4) taxonomies designed for domain-specific applications, and (5) taxonomies focused on Honey-X technologies. Additionally, several works provide descriptive overviews of the field without proposing formal taxonomic structures. In the following sections, we examine each category in detail, highlighting their key characteristics, contributions, and limitations. Table 3 provides a comparative summary of the taxonomies proposed across these categories.

Table 3. Summary of Taxonomies in General Cyber-Deception

Category	Papers	Core Characteristics	Pros	Cons
Military & Conceptual Strategy-Based	[9, 14, 22, 32, 33, 45, 68, 77]	Based on classical military doctrine. Divides deception into Simulation (showing the false) and Dissimulation (hiding the real). Focuses on high-level strategic goals.	- Conceptually grounded and intuitive. - Provides a clear, high-level starting point.	- Not native to the cyber domain. - Lacks technical granularity. - Categories can be ambiguous and overlapping.
Multi-Dimensional & Orthogonal	[31, 74]	Classifies techniques along multiple independent axes (e.g., goal, layer, deployment, kill-chain phase). Aims for a comprehensive and systematic framework.	- Provides a rich, structured, and holistic view. - Allows for systematic comparison and unification of concepts.	- Can be complex and difficult to apply in practice. - Assumed orthogonality of dimensions may not hold true. - May obscure high-level strategic intent.
Biologically-Inspired	[48, 52]	Uses a formal, tree-like structure based on biological analogies.	Novel perspective with a formal, logical structure.	- Completeness is not formally proven. - Categories are not mutually exclusive in practice. - Relies on unique characteristics; potentially restrictive definitions.
Niche & Domain-Specific	[6, 21, 28]	Tailored to specific sub-fields like game theory, Generative AI, or adversarial attacks on AI models.	High specificity and detail for the target domain.	- Not generalizable to the broader field of cyber-deception. - Sometimes uses the term "taxonomy" informally.
Honey-X Focused Enumerations	[54, 71]	Provides descriptive lists of honey-based techniques (honeypots, honeynets, honeytokens) rather than a formal classification structure.	- Simple and accessible for newcomers. - Clearly shows relationships between honey-based tools.	- Not a formal taxonomy; lacks classification dimensions. - Scope is limited to honey-based methods. - Definitions can be overlapping and incomplete.

4.1.1 Military Deception Adaptations. The most prevalent approach to classifying cyber-deception adapts classical deception theory from military doctrine. Taxonomies introduced by several works [9, 14, 22, 32, 33, 45, 68, 77] are predominantly founded on Whaley’s strategic framework [70], which distinguishes between two fundamental categories: *dissimulation*, hiding the real, and *simulation*, showing the false. Dissimulation focuses on concealing genuine assets and information through techniques such as *masking* (obscuring an object’s presence), *repackaging* (disguising an object’s true nature), and *dazzling* (confusing an observer to prevent accurate identification). Conversely, simulation

involves presenting false artifacts to mislead adversaries through techniques including *mimicking* (imitating a genuine object), *inventing* (creating novel false objects), and *decoying* (luring attackers toward false targets to divert attention from genuine assets).

Additional works within this category build upon similar strategic principles. For instance, [9] categorizes deception according to tactical objectives such as luring, misleading, or delaying attackers. Similarly, [14] proposes a framework based on the fundamental principles of creating ambiguity, distraction, and legitimacy to manipulate adversary perceptions. [33] offers an alternative but conceptually related model that frames deception within a matrix of *denial* (concealing facts) and *deception* (revealing fictions).

The principal strength of this approach lies in its foundation upon well-established deception theory, making it conceptually intuitive. The high-level distinction between *hiding* and *showing* provides a clear and accessible analytical framework. However, this clarity comes at a price. A critical limitation is that the framework originates outside the cyber domain and consequently struggles to accommodate dynamic or complex modern techniques such as MTD. Furthermore, the high level of abstraction inherent in these taxonomies often lacks the granularity required for direct technical implementation, and the proposed subtypes (e.g., repackaging versus mimicking) can exhibit ambiguity and conceptual overlap, complicating precise classification.

4.1.2 Multidimensional and Orthogonal Frameworks. To address the limitations of single-axis classification models, several works [31, 74] propose multidimensional taxonomies that classify deception techniques along multiple independent (orthogonal) dimensions.

Specifically, [31] introduces a formal classification system comprising four orthogonal dimensions: the *objective* of the deception (e.g., prevention, detection), the *asset* being protected (e.g., data, service), the *operational layer* (e.g., network, application), and the *deployment strategy* (e.g., built-in, isolated). In contrast, [74] employs a two-dimensional taxonomy that maps techniques against both the Cyber Kill Chain [34] phase they target and the deception layer at which they operate. This model successfully unifies diverse deception domains, including honeypots, honeytokens, and MTD, within a single coherent framework.

These taxonomies provide comprehensive and systematic classification, facilitating rigorous comparison and organization of diverse research contributions across the field. However, practical application presents challenges. The authors of [31] acknowledge difficulties in mapping specific publications to their proposed categories. The assumed orthogonality of dimensions may not hold consistently for real-world techniques, and the complexity introduced by multiple dimensions can obscure the high-level strategic intent that remains more transparent in military-based models.

4.1.3 Biologically Inspired Taxonomies. A third, more novel approach draws upon analogies from biological defensive mechanisms to construct a formal hierarchical classification. This category is exemplified by the framework introduced in [52] and subsequently adopted and extended in the [48] survey. This taxonomy structures deception as a tree with disjoint properties, where the first-level division is based on the treatment of private information. This distinction separates the field into *Crypsis*, which involves concealing information and is further subdivided to classify techniques including Perturbation, Obfuscation, and MTD, and *Mimesis*, which involves creating false realities and encompasses Honey-X techniques and Attacker Engagement strategies.

The primary advantage of this model lies in its formal logical structure, which provides clear systematic hierarchy. The biological analogy offers a novel and intellectually compelling conceptual lens for understanding cyber-deception. However, the taxonomy's completeness lacks formal verification. The framework derives from biological analogy rather than from principles that would guarantee comprehensive coverage of conceivable cyber-deception techniques.

Furthermore, as acknowledged by [48], the categories are not truly mutually exclusive, with certain techniques potentially fitting multiple branches.

4.1.4 Domain-Specific Frameworks. A fourth category of works [6, 21, 28] develops taxonomies tailored to specific sub-domains of cyber-deception rather than addressing the field comprehensively. Examples include frameworks for deception in attacker-defender game theory [28], applications of Generative AI for offensive and defensive deception [21], and classification of adversarial attacks against AI models [6].

These taxonomies offer high specificity and technical depth, providing relevant and actionable classification schemes for researchers operating within particular niches. However, their applicability is inherently limited by design, precluding generalization to the broader cyber-deception landscape.

4.1.5 Honey-X Focused Enumerations. Finally, two works [54, 71] provide descriptive overviews of honey-based techniques rather than formal structured taxonomies. In particular, [71] enumerates various "Honey-X" technologies, including honeypots, honeynets, and honeytokens. [54] extends this enumeration to emerging concepts such as honeyfarms, honeyclouds, and honeywebs.

The primary value of these enumerations lies in their accessibility. They serve as effective introductions for researchers new to the domain, clearly delineating the fundamental components of honey-based deception and their basic interrelations. The principal limitation is that these are not formal taxonomies. They lack systematic frameworks for classification and comparison, offering instead narrative summaries without rigorous analytical structure. Their scope is deliberately limited to honey-based mechanisms, excluding other major deception strategies such as MTD or obfuscation. Furthermore, as noted in [54], the definitions of enumerated categories exhibit overlap and incompleteness, and the works fail to abstract beyond individual examples to derive generalizable principles.

4.2 Moving Target Defense

Unlike the broader cyber-deception landscape, MTD literature demonstrates remarkable consensus in its classification approaches. The prevailing taxonomy addresses three fundamental questions:

- **What to move?** (Spatial Dimension) refers to the system property being changed, typically organized by system layer: Network (IP addresses, ports), Platform (operating systems), Application (code), and Data (format, encoding).
- **When to move?** (Temporal Dimension) describes the trigger mechanism for change, typically divided into: *Proactive* (time-based, predetermined intervals), *Reactive* (event-based, triggered by detection), and *Hybrid* (combining both approaches).
- **How to move?** (Mechanistic Dimension) refers to the underlying implementation technique: *Shuffling* (randomizing system properties), *Diversity* (deploying multiple variants), and *Redundancy* (maintaining multiple replicas).

While this framework provides a common conceptual vocabulary, surveyed papers apply it with varying degrees of completeness and specificity. Tab. 4 reveals three distinct organizational approaches, ranging from single-dimensional to comprehensive multidimensional technical frameworks.

4.2.1 Single-Dimension Taxonomies. Several papers [1, 23, 24, 26, 44, 51, 75, 76] adopt single-dimensional taxonomies that emphasize the system layers as their primary organizing principle. These taxonomies focus exclusively on the *What* dimension, creating a hierarchical catalog of MTD techniques organized by deployment layer: Network, Platform/Host,

Application, and Data. The *When* (temporal trigger) and *How* (implementation mechanism) dimensions are treated as secondary attributes describing techniques within each layer.

The principal advantage of this approach is its practical intuitiveness and accessibility. It provides defenders with a straightforward map for identifying which infrastructure components can be moved. However, this layer-centric focus risks underestimating critical temporal and operational aspects of defense, which are often equally important for effective deployment and analysis.

4.2.2 Two-Dimensional Taxonomies. A second group of works represents an evolution toward greater sophistication, constructing taxonomies along two of the three core dimensions. For instance, [66] builds its taxonomy on the *What* and *When* dimensions, while [40] emphasizes *What* and *How*. This approach offers more nuanced analysis than single-dimensional models by highlighting the crucial interdependencies between MTD aspects. However, these taxonomies remain inherently incomplete by design, as they omit the third dimension.

4.2.3 Three-Dimensional Taxonomies. The most comprehensive approaches are presented in papers [10, 13, 43, 57, 65], which employ the complete *What–When–How* framework, treating all three dimensions as co-equal classification axes. This approach provides the most robust and systematic model for describing and comparing MTD techniques across the full design space.

Despite the framework’s conceptual strengths, significant challenges persist across technical MTD taxonomies. Terminology remains inconsistent, with category definitions varying across papers.

Beyond these framework-based taxonomies, additional approaches exist with more limited scope. [55] provides a deep domain-specific taxonomy restricted to network-layer MTD techniques. Finally, [4, 67] do not provide a taxonomy of MTD techniques. Instead, they offer a broader classification scheme for organizing the entire field of study. These papers categorize the literature based on its contribution, creating buckets for topics like technology, architecture, performance metrics, testbeds, and evaluation strategies.

Table 4. Summary of MTD Taxonomy Groups

Category	Papers	Core Characteristics	Pros	Cons
Single-Dimension	[1, 23, 24, 26, 44, 51, 75, 76]	Primarily focuses on one dimension, overwhelmingly the "What" (system layer), or limits scope to one domain (e.g., networking).	Intuitive, practical, and aligns well with system architecture. Allows for deep analysis in a specific domain.	Can oversimplify techniques by downplaying other dimensions. Not comprehensive across the entire MTD space.
Two-Dimensions	[40, 66]	Creates a matrix by combining two of the three core dimensions (e.g., "What & When" or "What & How").	Offers a more sophisticated analysis than single-dimension models, highlighting key relationships.	Inherently incomplete by omitting the third critical dimension, which can limit its explanatory power.
Three-Dimensions	[10, 13, 43, 57, 65]	Utilizes the full, co-equal "What, When, How" framework to create a complete classification space.	The most comprehensive, robust, and systematic model for describing and comparing any MTD technique.	Can suffer from ambiguous definitions and overlapping categories, making clean classification challenging.

4.3 Honeypots

Among the four surveys addressing honeypots, only two [37, 72] propose formal taxonomies, each presenting a distinct classification approach.

4.3.1 Goal-Oriented and Application-Based Classification. The taxonomy presented in [72] classifies honeypots according to their intended function and high-level objectives within a security architecture. The framework employs a two-axis classification scheme. The first axis categorizes honeypots by their *Security Goals*: Prevention honeypots, which divert and deceive attackers away from genuine assets; Detection honeypots, which identify intrusions through monitoring system activity; Reaction honeypots, which function as sandboxes for post-incident investigation and analysis; and Research honeypots, which gather intelligence on attacker methodologies and tools. The second axis classifies honeypots by their *Application Goals*, yielding specialized categories such as Anti-spam honeypots (designed to capture unsolicited emails), DDoS honeypots (which detect and redirect distributed denial-of-service traffic), and Worm honeypots (built to capture and analyze automated malware).

The primary strength of this taxonomy is its pioneering establishment of a goal-oriented framework for honeypot classification. However, the framework exhibits significant limitations. It is largely conceptual and reflects network architectures prevalent in the early 2000s, potentially rendering it outdated in contemporary contexts. Additionally, the taxonomy was not validated through systematic literature analysis, limiting confidence in its completeness and applicability.

4.3.2 Technique and Theory-Based Classification. The second approach, outlined in [37], provides a more nuanced framework by establishing a clear distinction between high-level strategic principles and low-level implementable techniques. The taxonomy comprises two complementary levels: individual honeypots and honeynets. For individual honeypots, it enumerates practical techniques operators can implement, including Advanced Mimicking (creating highly believable decoy systems), Fake Cooperation (engaging with attackers in ostensibly helpful ways), and Honeytokens Bait (planting fake data or credentials that trigger alerts upon use). Concerning honeynets (networks of honeypots), they are categorized based on simulation, which involves creating attractive and believable decoys to present false assets, and dissimulation, which focuses on concealing the true nature of the network and monitoring infrastructure.

The primary advantage of this model is its explicit separation of high-level strategy from implementable techniques, facilitating both conceptual understanding and practical deployment. However, the framework exhibits notable limitations. The categories for individual honeypot techniques are not mutually exclusive: *Advanced Mimicking*, for instance, often functions as a prerequisite for other techniques rather than as an independent classification. Conversely, the high-level strategic framework for honeynets is described as too abstract and philosophical to provide direct, actionable guidance for developers and practitioners.

5 META SURVEY OF LITERATURE

Our analysis in Sec. 4 reveals a landscape characterized by a deep but narrow focus on established techniques, namely Honeypots and in particular Moving Target Defense. In this section, first, we explore the findings of the surveys related to cyber-deception in general, MTD, and Honeypots. Then, we highlight the critical gap in literature of the other deceptive techniques we identify in our taxonomy.

5.1 General Cyber-Deception Surveys

As a first step, we analyze surveys that address cyber-deception broadly, encompassing multiple deception techniques and approaches rather than focusing exclusively on a single technology. These works aim to provide comprehensive overviews of the field, though they vary significantly in scope, methodology, and analytical perspective. Our analysis examines the dominant research themes explored in these surveys, identifies the critical gaps and open challenges they consistently highlight, traces the evolution of research perspectives over time, and discusses the methodological limitations that constrain their contributions to the field. This examination shows a rapidly maturing discipline that has progressed from foundational conceptualization through formal optimization to contemporary concerns about AI integration and hybrid defense strategies, yet one that remains hindered by methodological inconsistencies and systematic coverage gaps. Tab. 5 summarizes the analyzed papers addressing cyber-deception in general, providing an overview of their scope, key contributions, and limitations.

5.1.1 Common Research Themes. Surveys addressing cyber-deception in general exhibit several dominant research themes that reflect both the field’s theoretical foundations and its practical evolution toward more sophisticated defensive capabilities.

Game-Theoretic Modeling. A significant research trend involves applying formal game-theoretic methods to model deception interactions rigorously. Works such as [30, 52, 77] review how Stackelberg games and signaling games can model attacker-defender dynamics, enabling the optimization of defensive strategies through mathematical frameworks. These approaches treat cyber-deception as a strategic decision problem, allowing defenders to compute optimal deception policies under assumptions about adversary objectives and capabilities.

Adaptive Deception Through Artificial Intelligence. Multiple surveys explore the integration of AI techniques to enhance deception capabilities and automation. Studies including [48, 77] examine how machine learning and deep learning can enable autonomous deception frameworks that improve both honeypot credibility and MTD strategy selection through data-driven adaptation. More recently, [6, 21] investigate the dual-use nature of Generative AI as both a tool enabling attackers to generate novel threats and as a mechanism for defenders to design sophisticated AI-driven deceptive defenses.

Cognitive and Behavioral Models. Moving beyond purely technical or game-theoretic approaches, some surveys explore psychological dimensions of deception. Notably, [28] introduces a novel framework integrating security games with cognitive models to create personalized deceptions tailored to individual attacker behavioral patterns and decision-making processes. This perspective recognizes that real adversaries exhibit cognitive biases and bounded rationality rather than the perfect rationality typically assumed in formal models.

Domain-Specific Applications. Some surveys focus on applying cyber-deception principles to specialized operational contexts. For instance, [22] examines deception techniques specifically designed to counter application-layer attacks such as SQL injection, while [45] develops a conceptual framework for deploying cyber-deception in Industrial Control Systems (ICS), where operational constraints and safety requirements necessitate adapted approaches. Additionally, [15] explores deception as a defense mechanism for Blockchain and Internet of Things (BIoT) ecosystems, addressing the unique challenges of decentralized and resource-constrained environments. Finally, [14, 68] bridge academic research and commercial practice by connecting theoretical concepts to industry frameworks such as MITRE Engage [17] (formerly MITRE Shield) and analyzing the commercial deception technology market, highlighting the translation of research into operational tools.

Table 5. Analysis of Surveys on Cyber Deception

ID	Scope & Focus	Finding	Future Works Proposed	Key Limitation
[52]	Game Theory (GT) models.	Formal GT-based taxonomy; links deception types to game models.	Deeper research into mimesis, advanced game models, and including human psychology.	Narrow scope (GT only); lacks practical effectiveness metrics.
[71]	Intro to Honeypots, -tokens, -nets.	Presents modern case studies like PhantomFS & IoT honeypots.	Enhance decoys with ML and deploy more advanced honeypots in cloud environments.	Narrow scope (misses MTD); shallow analysis; no methodology.
[77]	Deception via GT & Machine Learning (ML).	Multi-faceted taxonomy; finds GT provides strategy & ML enables it.	Balance realism vs. cost, study adverse effects of deception, and deeper GT-ML synergy.	No reproducible methodology; focus is on models, not empirical performance.
[22]	Deception for web applications.	Applies military deception taxonomy (simulation/dissimulation) to web security.	Create a testbed to evaluate attacker reactions (belief, suspicion, disbelief).	Over-relies on a single, non-technical taxonomy; no systematic methodology.
[33]	Strategic analysis of counter-deception.	Focuses on detecting adversary deception; introduces "deception chain."	Need for technical detection tools, CTI integration, and study of ethics.	Highly strategic, not technical; omits analysis of tools like MTD/Honey-X.
[25]	Systematic survey of deception planning models.	Feature-based comparison of 10 planning models.	Tool support for planning, trade-off analysis, and enhanced risk assessment.	Very small sample size (10 papers); focus on process, not outcome.
[31]	Systematic review of intrusion deception (excludes MTD/honeypots).	Four-dimensional taxonomy (Goal, Unit, Layer, Deployment); analyzes models & evaluation.	Automated deployment, evaluation of false-negative rates, and real-world reproducibility.	Deliberately strict and narrow scope; provides a deep vertical slice, not a broad overview.
[14]	Industry & market-oriented overview.	Maps deception to industry standards; provides deployment framework.	Need for more realistic, AI-driven, and scalable Honey-X solutions.	High-level and abstract; lacks technical implementation details.
[45]	Conceptual framework for deception in ICS.	Maps deception taxonomy onto 7 dimensions of control system security.	Use the proposed framework as a guide for more specific, detailed studies.	Highly theoretical; vague suggestions with no domain-specific context or validation.
[9]	Strategic guide to deception design process.	Introduces "deception story" (narrative) and "plot" (plan).	A need for a deception-specific language (DDSL) and rigorous test/evaluation methods.	Very theoretical; aimed at planners, not engineers; not a technical guide.
[28]	Research framework for personalized deception.	Combines Security Games with cognitive models to create behavior-aware defense.	Expand from lab games to real-world scenarios and apply to more deception types.	A conceptual proposal, not a survey; lacks large-scale validation.
[30]	GT applied to camouflage & MTD.	Deep critique of 6 papers; introduces differential privacy concept.	Need for more realistic (non-zero-sum) game models and analysis of operational costs.	Very narrow scope; no explicit methodology for paper selection.
[21]	Generative AI's impact on Cyber Kill Chain.	Frames GenAI as a dual-use tool for both advanced attacks and defenses.	Develop robust detectors for GenAI content and create dynamic, AI-driven deception environments.	High-level; weak link between specific GenAI threats and tailored defenses.
[54]	Survey of hybrid strategies.	Finds that truly integrated hybrid systems are a major, un-researched gap.	Need for better metrics and balancing cost vs. realism.	Weak methodology; finds very little "hybrid" research to actually survey.
[48]	Review of ML/DL for deception.	Finds ML/DL can create autonomous deception to increase attacker cost.	Design strategies with less overhead; determine optimal timing for MTD.	Purely qualitative; focus on ML overshadows the deception techniques.
[32]	Theoretical modeling of system states.	Proposes a formal state machine model for the deception lifecycle.	None mentioned.	Abstract; presents a conceptual model rather than a survey and does not discuss implementation or empirical validation.
[15]	Exploratory review for Blockchain-IoT (BloT).	Identifies deception as a promising but unexplored paradigm for BloT.	A broad call for research into all aspects of deception for BloT (AI, ethics, APTs).	High-level and speculative; discussion remains general and only loosely tied to BloT-specific security concerns.
[68]	High-level overview of host/network tech.	Connects research concepts to commercial products.	Deception at the data level, hybrid solutions, and better metrics.	Brief and shallow analysis; lacks a formal methodology.
[6]	Deception for Adversarial AI.	Frames deception as tool to exploit vulnerabilities in adversary ML models.	Generalization of defenses, explainability in AI (XAI), and meta-learning.	Lacks formal methodology; discusses general AI metrics, not deception-specific ones.
[74]	30-year review of honeypots & MTD.	2D taxonomy mapping techniques to kill chain phases & system layers.	Need for orchestration frameworks and deeper AI/ML integration for automation.	Weak methodology; omits the crucial human/psychological dimension.

5.1.2 Research Gaps and Open Challenges. When analyzed collectively, surveys addressing general cyber-deception reveal a consistent set of unresolved challenges that constrain the field’s theoretical maturation and practical adoption. **Insufficient Metrics and Evaluation Frameworks.** Multiple surveys [25, 31, 48, 54, 68, 74, 77] identify the lack of quantifiable and standardized metrics as a critical limitation. Beyond simple detection rates, researchers call for measures that capture attacker confusion, wasted adversary effort, defender cost–benefit ratios, and operational overhead. Some works propose more nuanced evaluation approaches: [22] advocates frameworks that monitor attacker psychological reactions (e.g., suspicion, frustration, disbelief) as primary success indicators, while [31] emphasizes reproducible real-world experiments that jointly evaluate detection performance and false negatives. Similarly, [54, 77] argue for moving from purely simulation-based studies to testbeds with realistic datasets and operational conditions. As we show in the MTD and honeypot literature (Secs. 5.2 and 5.3), this evaluation gap recurs as a cross-cutting challenge across the entire cyber-deception survey landscape.

Enhanced Integration with AI for Automation and Realism. Recent surveys increasingly emphasize the need for more dynamic, adaptive, and believable deception scenarios that leverage AI to respond in real-time to attacker behavior [15, 21]. Static deception systems risk detection by sophisticated adversaries who can identify patterns or inconsistencies; AI-driven adaptation promises to maintain deception credibility over extended engagements. Beyond realism, surveys such as [14, 31, 74] advocate for AI and machine learning to automate the deployment, management, and monitoring of deception systems, thereby reducing operational burden on security teams. A particularly compelling research direction discussed in [6] involves exploiting the inherent vulnerabilities of AI models through deception techniques specifically designed to counter AI-powered attacks. As adversaries increasingly employ machine learning for reconnaissance, vulnerability detection, and attack automation, defenses that can mislead or corrupt adversary AI systems represent a critical frontier in the ongoing security arms race.

Integrated Hybrid Systems. Several papers identify the future of cyber-deception as necessarily involving the combination of complementary techniques into unified defensive frameworks. [54] surveys existing hybrid systems and concludes that truly integrated implementations remain underdeveloped, with most research focusing on isolated techniques rather than their synergistic combination. Looking toward operational integration, [9] calls for tighter coupling of deception techniques with existing Command and Control (C2) frameworks and Security Operations Centers (SOCs) to enhance operational relevance. Deception cannot function as a standalone capability but must integrate seamlessly with incident response workflows, threat intelligence platforms, and security orchestration tools. At a higher architectural level, [74] highlights the need for orchestration platforms capable of coordinating multiple deception techniques across different system layers within a cohesive strategic framework. Such platforms would enable defenders to deploy layered deception strategies that simultaneously operate at network, application, and data levels, creating more robust defenses that are harder for adversaries to circumvent.

Human Factors and Psychological Dimensions. Many surveys critique the field’s predominant focus on technical mechanisms while neglecting the fundamentally human nature of adversarial interactions. [52, 77] note that game-theoretic models typically assume perfectly rational attackers, overlooking the reality of human cognitive biases, emotional responses, and bounded rationality. These works advocate for integrating psychological and behavioral models to create more realistic adversary representations that account for how humans actually make decisions under uncertainty and stress. Additionally, [33] addresses the under-researched area of counter-deception, the study of how defenders can detect when adversaries are deceiving them. This represents a critical gap in understanding bidirectional deception scenarios, where both attackers and defenders may employ deceptive tactics simultaneously.

Ethical Considerations. As deception techniques become increasingly proactive and sophisticated, potentially affecting third parties or operating in legally ambiguous spaces, [15, 33, 52] emphasize the urgent need to explore ethical implications and develop clear legal frameworks governing the deployment of deceptive defenses. Questions arise about the legitimacy of active deception tactics, the potential for collateral effects on innocent users, data privacy concerns in honeypot systems, and the appropriate boundaries between defensive and offensive cyber operations. Indeed, without addressing these ethical and legal dimensions, practical adoption of advanced deception techniques may face significant institutional and regulatory barriers.

5.1.3 Evolution of Research Perspectives Over Time. The analysis of the survey literature, when organized chronologically, reveals a clear three-phase evolution in the research community’s perspective on cyber-deception.

Phase 1: Foundational Concepts and Frameworks (2009-2016). The first phase is characterized by highly theoretical and conceptual work. Early papers focused on establishing foundational ideas and justifying deception as a viable defensive paradigm. Research from this period was primarily concerned with proposing conceptual frameworks by adapting military strategic concepts and formal logic to the cyber domain [9, 33], and suggesting, at a high level, how deception could be applied to new specialized contexts such as Industrial Control Systems [45].

Phase 2: Formalization and Optimization (2017-2021) The second phase is characterized by a concerted effort to make deception techniques, particularly MTD and honeypots, more rigorous, intelligent, and effective. This period began with attempts to create abstract models, such as state machines, to formally define the deception process and lifecycle [32]. The research perspective then quickly shifted to how it is possible to optimize deception. This optimization drive was dominated by two major parallel trends. First, a significant wave of research applied game-theoretic modeling to optimize defender strategies and model attacker rationality [28, 52], treating cyber-deception as a mathematical decision problem to solve with formal analysis and optimal solution computation. Second, a parallel trend focused on using AI and machine learning to create intelligent and autonomous deception frameworks [77] that could adapt dynamically to adversary behavior without constant human oversight. This phase also saw the field mature sufficiently to produce the first rigorous systematic literature reviews aimed at organizing the now substantial body of research on deception planning [25] and intrusion deception techniques [31].

Phase 3: Integration and New Arms Races (2022-Present) The third and current phase demonstrates a paradigm shift beyond optimizing isolated techniques toward integration, practical deployment, and responding to emerging threats. It is marked by mature retrospective surveys that consolidate the trends of the previous phase, such as comprehensive reviews of machine learning applications [48] and game-theoretic approaches [30]. Simultaneously, the current perspective is defined by three emergent themes that characterize contemporary research directions. First is hybridization and integration, exemplified by [54], which explicitly surveys hybrid strategies combining MTD, Honey-X techniques, and machine learning, even as it acknowledges that truly integrated implementations remain in their infancy. Second is a move toward maturity and practice, exemplified by papers such as [14] that focus on the commercial deception market and integration with industry standards like MITRE Shield. This practical orientation also manifests in expansion into new operational domains like BIoT [15], where deception techniques must be adapted to real-world constraints and requirements. Finally, the most contemporary papers [6, 21] confront the AI arms race, analyzing the dual-use nature of Generative AI as both a new threat requiring AI-powered defensive decoys and as an adversary itself, where deception can be weaponized to attack and mislead adversary machine learning models.

This clear progression, from isolated concepts, through a phase of optimizing specific techniques, to the current drive for integrated, adaptive, and AI-aware systems, demonstrates the field’s rapid maturation. This evolution underscores

the pressing need for a comprehensive framework, as proposed in this meta-survey, to unify these disparate but now converging research threads and provide a coherent map for the next phase of development.

5.1.4 Critical Limitations in Survey Literature. The analyzed survey literature on cyber-deception exhibits systematic limitations that constrain its reliability, comprehensiveness, and utility for both researchers and practitioners.

Methodological Rigor Deficiencies. A substantial portion of the surveyed works lack the methodological rigor expected in systematic literature reviews. Specifically, [9, 14, 21, 30, 32, 33, 45, 68, 71] present no formal explicitly declared methodology for literature selection and analysis, rendering their findings potentially subjective and their processes non-reproducible. Without transparency about search strategies, inclusion criteria, and selection processes, readers cannot assess comprehensiveness or replicate the analysis. Even when methodologies are present, they often exhibit significant weaknesses that compromise their reliability: [23] employs only a single keyword across two databases, severely limiting coverage and raising concerns about systematic bias in paper selection. Similarly, [54] claims to conduct a "systematic literature review" yet lacks clearly defined filtering criteria and reproducible selection processes. Additionally, [25], while systematic in its approach, analyzes only ten papers, severely limiting the generalizability of its findings and raising questions about whether such a small sample can adequately represent the broader research landscape. These methodological deficiencies highlight a fundamental tension in the survey literature: Many papers function more as expert reviews or position papers than as systematic, evidence-based syntheses.

Narrow Scope and Topical Bias. Our analysis reveals a significant and persistent publication bias within the cyber-deception survey literature. The field's-eye view provided by existing surveys is not holistic but rather heavily skewed toward a few mature techniques and enabling methodologies. This bias manifests in several interrelated ways. First, many surveys claiming to address cyber-deception comprehensively actually focus narrowly on the two most well-established and mature technique families: honeypots and MTD. The volume of research on these topics is so substantial that it has spawned dedicated survey categories (which we analyze separately in the following subsections). While this specialization provides valuable depth, it creates an implicit hierarchy where well-studied techniques receive disproportionate attention in surveys, while emerging or less-formalized techniques remain underexplored. Second, a large portion of the literature focuses on enabling methodologies rather than on deception techniques themselves. This includes numerous reviews dedicated to game theory [30, 52, 77], machine learning [6, 48, 77], and high-level strategic planning [9, 25]. This methodological emphasis means that even when various deception techniques are mentioned, they often serve as illustrative examples for formal models rather than being the primary subject of analysis. These biases create critical gaps in survey coverage that affect the field's development. Even systematic reviews with rigorous methodologies, such as [31], explicitly exclude certain technique categories to maintain a focused scope. Consequently, several important deception categories identified in our taxonomy remain almost entirely absent from survey literature or receive only superficial treatment. For example, tarpits and honey encryption are not the primary focus of any surveyed work, despite their distinct operational characteristics and use cases. Bias exploitation techniques, including breadcrumbing and strategic signaling, lack systematic review as a cohesive domain, though individual components occasionally appear in isolation (e.g., cognitive modeling in [28]). Even foundational techniques such as honey tokens lack dedicated surveys, instead being subsumed into broader umbrella categories like "Honey-X" [54] or "intrusion deception" [31], where they receive limited individual analysis. This work directly addresses these gaps by situating both heavily-surveyed and under-explored techniques within a unified analytical framework, providing a more balanced and comprehensive view of the cyber-deception landscape.

Theory-Practice Gap. A recurring limitation across the survey literature is the disconnect between theoretical models and practical implementation considerations. This gap manifests in several ways that limit the actionable value of survey findings for practitioners. Some works [6, 28, 32, 45] propose conceptually interesting frameworks that lack empirical validation or sufficient implementation details to assess real-world utility. While these theoretical contributions advance conceptual understanding, their practical applicability remains uncertain without evidence from deployment scenarios or realistic testbeds. Other surveys [9, 25, 33] explicitly target strategic or military audiences, emphasizing high-level planning and design considerations while deliberately omitting technical implementation details. This strategic perspective is valuable for decision-makers but leaves engineers and security practitioners without the technical guidance needed for actual deployment. Finally, some papers [52, 77] focus extensively on model validity and theoretical properties, such as game-theoretic equilibria, while neglecting practical considerations critical for practitioner adoption. These include performance overhead, system usability, deployment complexity, operational costs, integration requirements, and scalability limitations. Without addressing these practical factors, even theoretically sound approaches can face adoption barriers in production environments where resource constraints, legacy system compatibility, and operational continuity requirements dominate decision-making. This theory-practice gap reflects a broader challenge in security research: balancing academic rigor and theoretical advancement with the practical needs of organizations seeking to implement defensive technologies.

The Missing Metric: Data Quality and Granularity. While the literature widely critiques the absence of effectiveness and cost metrics, we uncover a more nuanced but equally critical blind spot: the lack of frameworks to quantify the granularity and quality of the collected threat intelligence. Cyber-deception systems, particularly honeypots, serve a dual purpose: They are not only defensive barriers but also high-fidelity sensors intended to extract Attacker TTPs (Tactics, Techniques, and Procedures). Research and surveys regarding advanced automated analysis of honeypot logs and Cyber Threat Intelligence extraction are limited. Similarly, existing evaluation frameworks almost exclusively focus on binary outcomes (*e.g.*, detection success vs. failure) or system impact (*e.g.*, latency overhead), neglecting the informational value of the engagement. There is currently no standardized metric to differentiate between the quality of data captured by a low-interaction simulation versus a high-interaction environment. Without metrics to measure this data granularity, it is impossible for practitioners to objectively calculate the Return on Investment (ROI) of deploying complex, resource-intensive high-interaction systems over simpler alternatives. Future evaluation frameworks must therefore move beyond simple success rates to include a metric that quantifies the depth, context, and actionability of the forensic data collected.

5.2 Moving Target Defense

Moving Target Defense has emerged as a prominent proactive security paradigm, attracting substantial research attention across diverse application domains. In this section, we analyze the existing MTD survey landscape, examining its primary contributions, the consensus on open challenges, and the methodological limitations. Tab. 6 summarizes these results.

5.2.1 Common Themes in Findings and Contributions. The analysis of the MTD-focused survey literature identifies three categories of contributions: the systematic organization of the field through taxonomies, the application of MTD principles to specialized domains, and the critical examination of evaluation methodologies.

Table 6. Analysis of MTD-Specific Survey Papers

Paper	Scope & Focus	Key Findings & Contributions	Identified Future Works	Key Limitation of the Survey
[55]	Brief review of network identity-based MTD.	New 7-attribute classification for network identity MTD.	None mentioned.	No methodology; very narrow scope; descriptive, not analytical.
[4]	MTD techniques for DDoS mitigation.	New framework mapping "what, how, when" to DDoS techniques and technologies (SDN, NFV, etc.).	Need for hybrid MTD frameworks, AI/ML integration, and standardized evaluation in real-world scenarios.	Purely descriptive; the proposed framework is not evaluated or validated; no implementation guide.
[57]	Survey of MTD in Software-Defined Networking (SDN).	New qualitative and quantitative metrics to evaluate MTDs.	Exploring hybrid surface shifting; study integration with real systems.	Weak methodology; proposed metrics ignore test environments and overhead cost.
[36]	Practical view of wired network MTD (explicitly excludes IoT/wireless).	Applies "What, How, When" & Kill Chain frameworks. Highlights lack of practical real-world implementations.	Better metrics, real-world testbeds, hardware-accelerated MTD, and hybrid strategies.	Deliberately limited scope (wired only); not fully reproducible methodology.
[65]	Survey of MTD with ML integration.	Analysis of MTD solutions combined with ML.	Integration with existing protection technologies.	No evaluation of ML-MTD techniques against traditional ones; consider honeypots as a kind of MTD.
[75]	Architectural perspective of MTD (OS, software, network layers).	Provides a formal classification of MTD strategies based on their architectural layer of deployment.	Need for mathematical models, risk/-cost analysis, SDN-based MTD, and lightweight MTD for IoT.	Weak, non-reproducible methodology (unspecified keywords, subjective filtering).
[66]	Intersection of MTD and Game Theory.	Dual-classification of MTD (spatial/temporal) and game models.	Balancing time/event strategies, models for unknown attacks, and scalable models for large networks.	No explicit methodology; focuses on models, not practical effectiveness; neglects attacker adaptation.
[76]	Broad survey of MTD techniques, evaluation, and applications.	Evaluation approaches to assess the effectiveness of MTD.	Call for MTD techniques integration with traditional security systems.	Talks about what to measure but doesn't provide standardized metrics.
[40]	MTD against APT.	Mapping of MTD techniques to different steps of the CKC.	Evaluation of security level of MTD techniques using graphical security models.	Consider only IP randomization; other MTD techniques are not evaluated.
[67]	Systematic Mapping Study of MTD in Cloud Computing.	Maps research by area & MTD type. Finds most work is on platform/network.	Need for cloud-specific MTD theories, unified evaluation frameworks, and multi-layer MTD.	Weak search string (might miss non-"cloud" papers); risk of manual classification bias.
[23]	MTD for Cyber-Physical Systems (CPS).	Maps specific MTD techniques to real-world attacks against CPS.	Implement MTD in real CPS; create hybrid and new CPS-tailored methods.	No methodology; limited to CPS.
[44]	Broad, comprehensive survey of MTD.	Organizes MTD into Strategy Formulation, Transformation Mechanisms, and Effectiveness Evaluation.	Research optimal dynamic timing and unified quantitative criteria.	Complete absence of a structured methodology; future directions are abstract.
[51]	MTD for DoS mitigation in Cloud/SDN environments.	Categorizes MTD for DoS (IP shuffling, VM migration) and surveys testbeds/-metrics used for evaluation.	Optimize MTD timing, integrate with IDS/IPS, address gap in AI/ML integration, expand to SD-WAN.	No explicit methodology; implies selection is based on author relevance.
[10]	High-level survey of MTD state of the art.	Proposes a "cube model" (Layer, Functionality, Area) to organize MTD. Categorizes evaluation into 4 types.	Improve practicality, develop hybrid MTD, better comparison metrics, and apply game theory/SDN.	No explicit methodology; model is high-level; not a systematic review.
[12]	Book chapter on MTD for memory protection (ASLR/DSLR).	Evolution from static to dynamic ASLR/DSLR, using Chameleon and SALADS as case studies.	More adaptive ASLR/DSLR, feedback control mechanisms for cost, and better binary instrumentation.	No formal methodology (book chapter); a very narrow, deep technical review.
[43]	Proposal for an MTD visualization engine.	Adapts "What, How, When" and a 4-stage workflow (Detect, Respond, Reconfigure, Restore) to its tool.	None mentioned for the MTD field (only for their tool).	It's a systems paper with a supportive literature review. No methodology.
[50]	Systematic review of MTD for IoT.	Entropy-related metrics for evaluation; lack of real-world deployment for many techniques.	Call for real-world deployment; need to use stronger cryptographic solutions.	Papers covered up to July 2020; the entropy metrics are qualitative.
[24]	Quantification of MTD via an expert opinion survey.	Provides a coarse ordering of 24 MTD techniques by cost/effectiveness.	Calibrate survey results against empirical/testbed data.	Respondent fatigue; potential expert bias; lack of threat models.
[1]	Systematic review on SDN.	Exploration of MTD techniques to improve SDN security solutions.	Call for better overhead trade-off for MTD techniques in SDN context.	Narrow scope: focus on SDN.
[73]	Broad review of Smart Grid security; includes a section on MTD.	Identifies MTD as a key defense for smart grids, linking cyber techniques to physical consequences.	None mentioned for MTD.	Weak methodology (just "last 5 years"); MTD is only a small part of the review; limited scope.
[26]	Part review, part proposal for an MTD service framework.	New framework with an optimal user-server mapping mechanism.	None mentioned (discussion of gaps, but not as future work).	Framework relies on static, known values for risk/vulnerability.
[13]	Comprehensive survey of MTD as a proactive defense.	Uses the Shuffling, Diversity, Redundancy (SDR) framework. Explicitly distinguishes MTD from deception.	Interplay with other defenses, need for realistic testbeds, optimal hybrid MTD, and comprehensive metrics.	No explicit methodology; uneven analysis (e.g., 'what to move' is weak).

Development of Taxonomies. A significant portion of the surveyed literature dedicates substantial effort to creating structured taxonomies that manage the field’s inherent complexity, aiming to bring order to the diverse landscape of MTD techniques and approaches.

Domain Specific Application. The MTD survey literature exhibits substantial fragmentation, with numerous papers providing deep-dive analyses of specific environments. Cloud computing represents a major focus area, exemplified by systematic mapping studies such as [67] analyzing MTD applications in cloud environments, and specialized surveys like [51] examining DoS mitigation specifically. This pattern repeats across Software-Defined Networking (SDN) [1, 57] and particularly in Internet of Things (IoT) and Cyber-Physical Systems (CPS) [23, 50, 73], where surveys map MTD techniques across all system layers, from network infrastructure down to controller and physical components. This domain-specific focus extends to highly specialized technical problems, including dedicated reviews of DDoS mitigation [4, 51], network identity management [55], and memory protection mechanisms (ASLR/DSLR) [12]. While this specialization demonstrates the field’s maturation in applying general principles to practical challenges, it simultaneously contributes to the siloed nature of MTD literature, making comprehensive understanding difficult for researchers outside specific subdomains.

Evaluation Methodologies. Although most surveys identify evaluation as a critical research gap, few make the analysis of evaluation practices itself a primary contribution. Several surveys provide valuable overviews of the evaluation landscape, systematically categorizing testbed approaches (simulation-based, hardware platforms, cloud testbeds) and evaluation metrics (performance overhead, security effectiveness) [10, 44, 50, 51, 57, 76]. The most distinctive contribution in this category is [24], which departs from traditional literature surveys by implementing a novel expert survey methodology, proposing a framework to quantify and rank MTD techniques based on aggregated expert assessments of cost and effectiveness.

5.2.2 Findings and Open Challenges. Despite their fragmentation across domains, MTD-focused surveys demonstrate remarkable consensus in identifying open problems.

Standardized Evaluation and Cost Analysis. Nearly all MTD surveys identify the absence of standardized evaluation frameworks and rigorous cost–benefit analysis as a central gap [10, 13, 36, 44, 67, 76]. Echoing the broader evaluation challenges discussed in Sec. 5.1, current studies rely largely on ad-hoc, simulation-based experiments that make it difficult to compare techniques or demonstrate practical value. In addition to security effectiveness, multiple works stress the need for models that capture trade-offs between security gains and operational costs, including performance overhead, economic impact, and deployment risks [24, 36, 66, 75], so that practitioners can make evidence-based adoption decisions.

Real-World Applications. The call for standardized evaluation directly connects to the second major theme: the persistent need to transition MTD from theoretical promise to practical deployment. Several works critique the over-reliance on simulation environments and advocate for real-world applications and realistic large-scale testbeds for proper validation [4, 13, 36, 50, 57]. This emphasis on practicality extends to implementation considerations. For instance, [36] identifies hardware-accelerated MTD (leveraging SmartNICs) as a critical research direction for mitigating the performance overhead that currently impedes production adoption. Similarly, [1, 57] highlight scalability challenges that must be addressed for real-world SDN deployments.

Hybridization and Integration. The literature consistently frames MTD’s future as necessarily hybrid and integrated. Surveys identify the need for research into hybrid MTD strategies that combine multiple defensive techniques, such as shuffling and diversity, to create more resilient and multi-layered defenses [4, 10, 13, 36, 40, 57]. Beyond combining

MTD techniques internally, researchers also emphasize understanding the interplay between MTD and complementary defensive mechanisms. For example, integrating MTD with traditional intrusion detection and prevention systems (IDS/IPS) could enable more sophisticated cue-driven adaptation [13, 51].

Intelligent and Lightweight MTD. Finally, surveys consistently call for MTD to become simultaneously smarter and more resource-efficient. A dominant theme is the integration of Artificial Intelligence and Machine Learning to enable more adaptive, autonomous, and real-time MTD strategies capable of selecting optimal defensive moves without human intervention [1, 4, 51, 65]. In parallel, given the proliferation of resource-constrained computing environments, researchers urgently advocate for developing lightweight MTD techniques suitable for IoT, CPS, and smart grid deployments [1, 23, 65, 75]. These domains present unique challenges: They require effective security but operate under severe constraints in processing power, memory, and energy consumption.

5.2.3 The Evolution of MTD Survey Perspectives over Time. The analysis of MTD-focused surveys shows a distinct and logical evolution in the literature.

Phase 1: Foundational Organization (2014-2020). This initial phase centered on defining and structuring the emerging MTD field. Unlike the broader deception landscape, this period achieved rapid taxonomic consensus around three fundamental questions: What to move, When to move, and How to move. Surveys from this phase built this framework with varying levels of completeness. Many foundational works offered practical, single-dimension taxonomies focused primarily on the What dimension, cataloging techniques by system layer [23, 24, 44, 75]. Concurrently, more comprehensive surveys introduced the complete three-dimensional What-When-How framework [10, 13, 57], establishing the organizational structure that would guide subsequent research.

Phase 2: Domain Specialization (2020-Present). The second and current phase shifts the research perspective toward understanding how to apply MTD to specific domains. This led to a wave of surveys focused on narrow and practical environments, creating the highly fragmented landscape we see today. We find works dedicated to Cloud Computing [51, 67], Software-Defined Networking (SDN) [1, 40, 51], IoT and CPS [50, 73], and game theory applications [66].

5.2.4 Recurring Limitations in the Survey Literature. The MTD survey literature itself exhibits several limitations that constrain its utility.

Lack of Rigorous Methodology. Most MTD surveys are qualitative reviews that present findings without explicit, systematic, or reproducible paper-selection procedures. Many do not specify search keywords, databases, or inclusion/exclusion criteria [10, 13, 44, 51, 55, 57, 65, 66, 73, 75, 76], and even works that attempt to define a methodology often acknowledge overly restrictive search strings or other limitations [36, 67]. This mirrors the methodological deficiencies observed in general cyber-deception surveys (Sec. 5.1) and weakens confidence in the completeness and neutrality of their findings.

Highly Fragmented Scope. The MTD survey landscape is dominated by narrowly scoped papers. Most reviews target specific application domains such as cloud computing [51, 67], SDN [1, 57], or CPS and smart grids [23, 73], or focus on particular threat models and techniques, including DDoS mitigation [4], network identity management [55], memory protection [12], or game-theoretic approaches [66]. Even surveys with systematic methodologies, such as [50], typically analyze relatively small paper sets, reflecting the narrow boundaries of their chosen focus areas and making it difficult to obtain a holistic view of MTD from any single source.

Non-Traditional Survey Papers. The landscape is further complicated by papers that deviate from traditional survey formats. For instance, [26, 43] are primarily proposals for novel MTD systems and frameworks: their literature

review components serve to motivate design choices rather than provide exhaustive, unbiased analyses of the field. Similarly, [24] represents an expert opinion survey rather than a literature review, with findings derived from aggregated practitioner perspectives rather than systematic synthesis of published research.

5.3 Honeypots

Unlike the MTD literature, which saw a rapid consensus-driven organization, honeypot surveys are fragmented across time and purpose, as we summarize in Tab. 7.

Table 7. Analysis of Honeypot-Focused Survey Papers

Paper	Scope & Focus	Finding	Future Works	Key Limitation of the Survey
[37]	Qualitative survey on honeypot/honey-net.	Proposes two taxonomies: one for honeypots (mimicking) and one for honeynets (Simulation/Dissimulation).	AI/ML for adaptive honeynets; need for robust evaluation metrics.	No formal review methodology reported; narrative synthesis of prior work; proposed taxonomies have overlapping categories and limited guidance on boundaries, which may hinder practical application.
[72]	Foundational, conceptual review of honeypots as an active defense system.	Provides a high-level conceptual model (data capture/control) and a goal-based classification.	(Historical) Need for integration (single device), virtualization, and distribution (honeynets).	No formal review methodology; high-level overview rather than systematic survey; predates virtualization, containerization, and cloud-native deployments (2003), limiting coverage of later developments.
[59]	SLR using PRISMA on honeypots for data security.	Categorizes modern use by domain (IoT, cloud, ICS) and integration with AI, CTI, and MITRE ATT&CK.	Maintenance/resource costs; risk of recognition; AI integration; need for evaluation methods.	The synthesis of the 38 papers remains relatively high-level, with limited comparative analysis of individual techniques.
[2]	Survey and assessment of wireless honeypots (pre-WPA2 era).	Classifies wireless honeypots by interaction level & deception method.	(Implied) Need for effectiveness metrics, performance impact analysis, and attacker behavior studies.	Relies on secondary reports rather than primary empirical evaluation; no formal review methodology documented; focuses on pre-WPA2 wireless environments (2009), which limits applicability to current WLAN deployments.

5.3.1 Common Themes in Findings and Contributions. Honeypot survey literature primarily contributes classification schemes and, more recently, identifies integration opportunities with modern security systems.

Definition of Taxonomies. Early surveys focused on establishing conceptual clarity through taxonomic frameworks. The foundational work of [72], published in 2003, established an early conceptual model that distinguished honeypots based on their data capture capabilities and control mechanisms (how much attacker freedom they permit), while classifying them by high-level goals such as research, detection, or threat intelligence gathering. Similarly, the 2009 survey [2] adapted these foundational concepts to wireless environments, categorizing wireless honeypots by interaction level and specific deception methods unique to wireless contexts, such as spawning fake access points, emulating network traffic patterns, and simulating authentication protocols.

Modernization and Integration. More modern surveys shift from defining honeypots to enhancing them. In 2024, [37] proposed detailed taxonomies for performance-improving techniques that extend honeypot capabilities beyond basic deception. These include Advanced Mimicking (techniques for creating more realistic system behaviors and responses), Fake Cooperation (methods for prolonging attacker engagement by simulating successful exploits), and Honeytaken Bait (strategic placement of attractive but fake credentials, data, or resources to lure attackers). The systematic review presented in [59] takes a different modernization approach by categorizing contemporary honeypot applications according to two dimensions: operational domain and integration context with complementary security technologies.

This integration perspective is particularly significant, as [59] documents how honeypots are increasingly embedded within AI/ML systems for automated threat analysis, Cyber Threat Intelligence platforms for indicator sharing and correlation, and the MITRE ATT&CK [16] framework for attack pattern mapping and threat hunting.

5.3.2 Future Directions. Despite the 20-year span of the surveyed literature, there is a remarkable and persistent consensus on the field’s most critical research gaps.

Lack of Standardized Metrics. The absence of standardized evaluation frameworks dominates the literature from 2003 to 2024, appearing as the primary future-work recommendation across nearly all surveyed papers. The critique in [2] regarding missing metrics for effectiveness is echoed almost identically in recent works [37, 59], which call for comprehensive measures of deception realism, intelligence value, operational cost, and attacker dwell time. This long-standing gap reflects the broader evaluation challenges discussed in Sec. 5.1 and continues to hinder comparative studies and evidence-based honeypot deployment decisions.

Need for Intelligent Adaptation. Modern papers [37, 59] identify intelligent adaptation as the second major gap, reflecting contemporary interest in autonomous security systems. Both papers advocate for AI integration to create adaptive honeynets that can learn from attacker interactions and dynamically improve deception strategies without manual reconfiguration. Additionally, [59] identifies a set of practical, real-world challenges that complicate honeypot deployment beyond technical capabilities. These include high maintenance costs for keeping deception environments current with legitimate system updates and patches, significant resource demands of high-interaction honeypots that require full operating systems and application stacks, the specialized expertise needed to configure realistic deception scenarios, and the persistent risk of attacker recognition and evasion through fingerprinting techniques, behavioral analysis, or community knowledge sharing.

5.3.3 Evolution of Honeypot Surveys. The analysis of these honeypot surveys, spanning over two decades, illustrates a clear two-phase evolution.

Phase 1: Foundational Concepts (2003-2009). Early works [2, 72] focused on defining the emerging field during a period when honeypots transitioned from niche security tools to recognized research instruments. These papers established conceptual models that distinguished honeypots from intrusion detection systems and firewalls, defined basic classifications including goal-based taxonomies (research vs. production honeypots) and interaction-level distinctions (low, medium, high interaction), and began systematically applying honeypot concepts to emerging domains like wireless networks and mobile systems.

Phase 2: Modern Integration and Evaluation Crisis (2020-Present). Current works [37, 59] shift focus from definition to enhancement, operating in an environment where honeypots are established but require modernization to address sophisticated threats. This phase emphasizes improved believability through advanced mimicking techniques that defeat fingerprinting attempts [37] and integration with modern security ecosystems including AI-driven threat analysis and MITRE ATT&CK framework mapping for standardized attack pattern documentation [59]. However, this modern phase is still constrained by the evaluation gap noted above: recent papers [37, 59] repeat the same calls for standardized metrics already identified in 2009 [2], indicating that the field still lacks robust methods for demonstrating honeypot effectiveness.

5.3.4 Recurring Limitations. Honeypot survey literature exhibits two primary limitations that constrain its impact and reliability.

Methodology Weakness. The most significant limitation is insufficient methodological rigor in literature review processes. Three of the four surveys [2, 37, 72] are qualitative reviews lacking formal, systematic, or reproducible paper-selection methods, which prevents independent replication and makes it difficult to assess potential gaps in coverage. In contrast, [59] adopts a PRISMA-based methodology with documented search strings, multi-database queries, and explicit selection criteria, but its analysis remains largely descriptive and surface-level. This combination of informal methods and shallow synthesis mirrors the broader survey-methodology issues observed for general cyber-deception and MTD (Secs. 5.1 and 5.2).

Obsolescence and Flawed Contributions. Furthermore, the existing honeypot survey literature suffers from temporal and methodological limitations that reduce its current practical utility. Foundational papers from 2003 [72] and 2009 [2] remain important historically, but their taxonomies predate virtualization, containerization, and cloud-native honeypots, and their deployment recommendations assume pre-cloud infrastructure models. As a result, they provide only limited guidance for contemporary environments. More recent work such as [37] faces different challenges. Its central contribution—the taxonomy of honeypot enhancement techniques—contains overlapping and ambiguously defined categories, so that several techniques can naturally fall into multiple classes without clear assignment criteria, and the high level of abstraction requires substantial interpretation before the taxonomy can be applied in practice. Taken together, these limitations leave practitioners with little up-to-date, methodologically robust survey guidance for understanding the current state of honeypot research and making evidence-based deployment decisions.

6 OUR TAXONOMY

6.1 Gaps in Existing Taxonomies

Despite two decades of research producing numerous classification schemes for cyber-deception, the analysis in Sec. 4 reveals that no existing taxonomy provides the comprehensive, unified, and operationally actionable framework necessary to guide both research and practice. The fragmentation of the survey literature has produced taxonomies that, while individually valuable, collectively fail to address four critical gaps that hinder the field’s maturation.

6.1.1 Incomplete Technical Coverage. The most significant gap is the systematic exclusion or superficial treatment of important deception techniques. Existing surveys exhibit strong bias toward two mature technique families, honeypots and MTD, while ignoring other approaches with distinct operational characteristics and proven practical value.

Tarpits, despite their effectiveness in resource exhaustion attacks and fundamentally different operational principles from honeypots, receive no dedicated coverage in any surveyed work and appear only as brief mentions in broader discussions. Honey encryption remains almost entirely absent from systematic surveys, despite operating on principles distinct from both honeypots and MTD. Honey patches similarly lack coverage despite their unique approach of strategically managing vulnerability disclosure and remediation. Most critically, bias exploitation receive minimal systematic attention despite their growing importance in sophisticated defense strategies. When mentioned at all, these approaches appear as superficial observations. The few works that address cognitive dimensions [28, 33] treat them as narrow research questions rather than as fundamental deception categories.

6.1.2 Incompatible Classification Schemes. Beyond incomplete coverage, existing taxonomies employ fundamentally incompatible organizational principles that prevent integration and comparison across the literature.

The most prominent incompatibility exists between strategy-based taxonomies adapted from military doctrine [9, 14, 22, 32, 33, 45, 68, 77] and technically-oriented frameworks organized by system implementation [31, 74, 75]. Military-derived taxonomies classify deception according to strategic intent, providing intuitive high-level categories but struggling to accommodate dynamic techniques like MTD that do not map cleanly onto these conceptual divisions. Conversely, technical taxonomies organized by deployment layer excel at describing implementation details but obscure strategic objectives. A honeypot and a tarpit might both operate at the network layer, yet serve fundamentally different defensive purposes.

6.1.3 Absence of Operational Guidance. While existing taxonomies successfully classify techniques according to various criteria, they largely fail to provide the operational guidance necessary for practitioners making deployment decisions. This gap between taxonomic classification and practical action manifests in two critical ways. First, many taxonomies employ ad-hoc, inconsistent, or vaguely defined goal categories that provide little actionable direction. Terms like "prevention," "detection," "monitoring," "distraction," and "confusion" appear across surveys with varying definitions and scope. Without standardized goal definitions, practitioners cannot reliably determine which techniques serve their particular security needs. Most critically, previous taxonomies exist in isolation from the operational frameworks and standards that guide real-world security practice. Defenders working within established security operations workflows, mapped to frameworks like the MITRE ATT&CK matrix [16] for threat modeling or the Cyber Kill Chain [34] for understanding attack progression, find little guidance about how taxonomic classifications translate into operational deployment decisions within these existing frameworks.

6.1.4 Domain-Specific Models Without Cross-Domain Integration. The cyber-deception literature has evolved into distinct domain-specific research communities (e.g., cloud security, IoT, and SDN) each developing specialized taxonomies and vocabulary optimized for their particular context. While this specialization has enabled deep technical expertise within domains, it has simultaneously created silos that impede knowledge transfer and prevent the development of unified conceptual frameworks applicable across operational contexts. Surveys targeting specific domains [23, 45, 50, 51, 67, 73] rarely reference or build upon classification schemes developed in other domains, even when addressing fundamentally similar deception mechanisms. Each domain has reinvented taxonomic wheels rather than adapting and extending cross-domain frameworks. This siloing manifests most visibly in terminological inconsistency: The same deception technique may be described using entirely different vocabulary depending on the domain literature in which it appears. Perhaps most problematically, domain-specific silos prevent researchers from identifying generalizable principles that transcend particular operational contexts. A deception technique developed for cloud environments might prove equally applicable to IoT systems if expressed at an appropriate level of abstraction, yet domain-specific framing obscures this potential.

6.2 A Three-Dimensional Unified Taxonomy

To address these systematic gaps, we propose a new comprehensive taxonomy grounded in three core design principles: dimensional separation to avoid conflating distinct aspects of deception, operational pragmatism to bridge research and practice, and systematic completeness to ensure comprehensive coverage of the technique landscape. Our taxonomy employs three dimensions that separate concerns traditionally conflated in existing frameworks: the Deception Technique dimension, the Deception Goal dimension, and the Deception Placement dimension.

The Core Technique Dimension. We establish deception technique as the primary organizing dimension, creating the taxonomic backbone based on operational mechanics, how techniques actually function, rather than their strategic

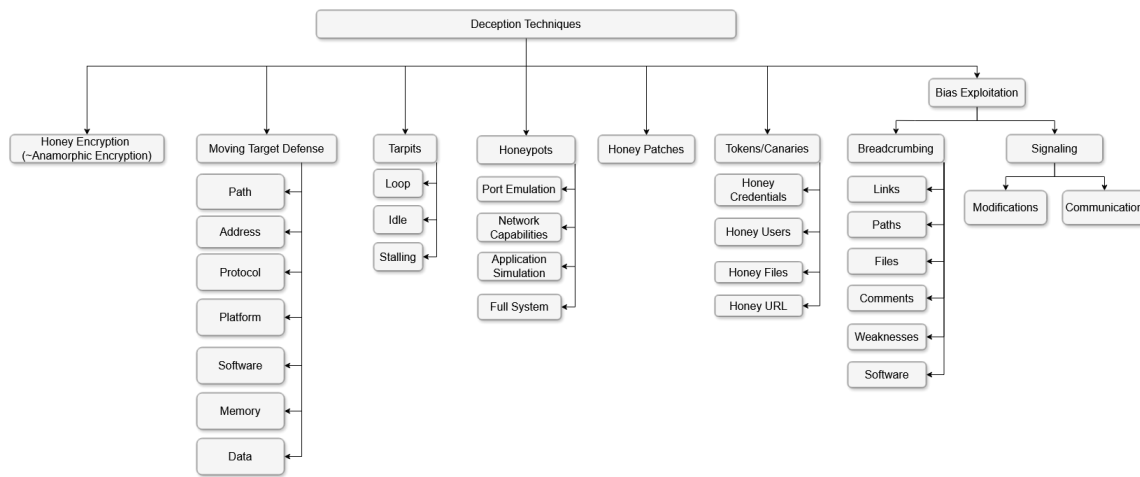


Fig. 2. Cyber Defense Deception Techniques Taxonomy.

intent or deployment context. This choice reflects our observation that strategic goals and deployment contexts are better treated as labels or attributes that can be applied to techniques rather than as primary classification criteria. A honeypot remains fundamentally a honeypot whether deployed for intelligence gathering or threat detection, whether running in a cloud environment or on physical hardware. Its essential nature is defined by its operational mechanism, not by why it's deployed or where it runs. By making technique the core dimension, we create a stable classification scheme that remains valid across varying strategic objectives and deployment contexts.

Orthogonal Labeling Dimensions. The Goal and Placement dimensions function as orthogonal labeling systems that contextualize techniques without constraining their classification. Any technique can be labeled with multiple goals it serves and multiple placement locations where it can be deployed, without affecting its position in the core technique taxonomy.

This three-dimensional structure provides multiple advantages. First, it allows practitioners to navigate the taxonomy from multiple entry points: Security teams can start with a goal and identify candidate techniques, or start with a deployment context and discover applicable techniques, or start with a technique they are familiar with and understand its full range of applications. Second, it facilitates gap analysis: Defenders can examine which goals are addressed by their current deception deployments and which placements lack coverage, identifying gaps in their defensive posture. Third, our taxonomy supports progress toward addressing the evaluation crisis: By providing categorical structure and standardized vocabulary, our taxonomy creates the foundation necessary for community-wide comparative benchmarking efforts that can enable meaningful effectiveness comparison and scientific maturation of the field.

6.3 The Core Dimension: Deception Techniques

The core of our taxonomy categorizes cyber-deception into distinct technical approaches, as illustrated in Fig. 2. We hierarchically organize techniques from major families, distinguished by fundamentally different operational principles, down to implementation-level variations. Specifically, we explicitly exclude ambiguous terms like *decoy*, which is used inconsistently in literature, and overarching terms like *Honey-X*, which conflate dissimilar mechanisms. Instead, we classify techniques based on their operational definitions.

6.3.1 Honeypots. We classify honeypots by implementation depth, the extent of functionality provided to attackers, rather than traditional low/medium/high interaction terminology that lacks clear definitions. At the foundational level, we define *Port Emulation* (e.g., reactive telescopes), where systems show only open ports and fail any further interaction. This progresses to *Network Capabilities*, which includes protocol emulation without application logic, followed by *Application Simulation*, where specific services are simulated. Finally, *Full System* implementations allow for the complete execution and logging of an attack within a high-fidelity environment. This classification enables practitioners to select appropriate fidelity levels based on their intelligence gathering requirements and resource constraints while maintaining clear taxonomic boundaries.

6.3.2 Moving Target Defense. We focus classification on what asset is being mutated rather than when to move or how to move. This reflects our observation that temporal strategies (proactive vs. reactive triggers) and implementation methods (shuffling vs. redundancy) are orthogonal attributes applicable to almost any MTD technique. By focusing on the specific asset being protected, we provide technique-centered abstraction that avoids mixed-dimension complexity. In particular, we categorize these techniques by the system layer they manipulate. The *Network Layer* covers communication surface changes, including Address (IP, Port, MAC hopping) and Path (routing, flow table randomization). The *Platform Layer* includes OS rotation and virtualization-based migration. At the *Software/Application Layer*, techniques cover software diversification and middleware diversity. The *Runtime/System Layer* focuses on memory and process behavior, such as ASLR, ISR, and dynamic runtime environment transformation. Finally, the *Data Layer* involves data randomization, including changes to format, syntax, and representation.

6.3.3 Distinct Specialized Techniques. We elevate previously marginalized techniques to first-class taxonomic status, addressing the incomplete coverage gap. Rather than subsuming these under broader *Honey-X* categories or omitting them entirely, we provide a dedicated classification that acknowledges their distinct operational principles.

Tarpits. Unlike honeypots that gather intelligence through realistic engagement, tarpits are resource-consumption traps designed to waste attacker time with minimal defender investment. We classify their operation into three distinct mechanisms. First, *Loops* involve the deliberate creation of routing or protocol cycles to trap automated scanners in endless repetitions. Second, *persistence* mechanisms prevent connection termination, by ignoring close requests, remaining idle, or resending keep-alive packets, the system forces the attacker’s connection to remain open indefinitely. Finally, *Stalling* exploits protocol standards by maximizing response latency at every step, technically maintaining compliance while reducing the speed of automated attacks to a crawl.

Honeytokens (Canaries). Honeytokens are passive deceptive assets, such as fake credentials, database entries, API keys, or documents, embedded directly within real production systems. They are distinct from honeypots (which are separate environments) as they serve a purely detection role: Since they have no legitimate business function, any interaction with them is by definition an anomaly that triggers a high-fidelity alert.

Honey Encryption. This technique is distinct because it targets the confidentiality medium rather than creating fake assets. Honey encryption generates plausible-looking but fake data when decrypted with an incorrect password. This denies the attacker the ability to know if they have succeeded, protecting the real data even after exfiltration.

Honey Patches. Unlike honeypots that simulate vulnerable systems, Honey Patches apply a fake fix to a real vulnerability in a production system. The patch appears to rectify the flaw to standard checks, but internally redirects any exploit attempts against that specific vulnerability to a contained environment or logger, allowing the defender to monitor the attack safely on a live system.

6.3.4 Bias Exploitation and Psychological Deception. A major contribution of our taxonomy is establishing Bias Exploitation as a fundamental top-level category alongside technical mechanisms. Existing taxonomies, when addressing cognitive dimensions at all, treat them as secondary attributes or marginal considerations. We argue this is fundamentally misguided. Cyber attacks involve human decision-makers (even when using automated tools), and human cognition is subject to biases and manipulability that technical deception alone cannot fully exploit. By establishing psychological deception as a first-class category with detailed subcategories, we provide a systematic organization for a previously fragmented area and signal that effective defense requires integrating both technical and psychological approaches. We divide this category into *Breadcrumbs* and *Signaling*.

Breadcrumbs. It is a passive, static approach involving the strategic placement of assets, such as files, comments, or links to create a path that manipulates the attacker’s perspective. While honeypots are strictly fake entities designed to trigger an alert upon interaction, breadcrumbs can utilize both fake and real assets, including valid files, actual software comments, or existing system configurations, to build credibility. Breadcrumbs do not need to be monitored and trigger alarms. Furthermore, while a honeypot is often a standalone trap, breadcrumbs are typically arranged in a chain to guide the attacker away from high-value targets and towards a desired destination (such as a honeypot).

Signaling. Signaling is a reactive form of bias exploitation, where the system responds dynamically to the attacker’s presence. Unlike static defenses, signaling alters the environment during an intrusion to shape the attacker’s decision-making. We categorize signaling into *Communication* and *Modification*. The first approach involves direct interaction with the attacker: The system sends explicit messages, such as administrative warnings or fabricated error logs, intended to provoke an immediate reaction (*e.g.*, causing the attacker to panic, retreat, or rush their execution). The Modification approach involves indirect manipulation requiring attacker deduction. The system alters its state in real-time, for example, changing file permissions, hiding previously visible directories, or subtly corrupting data content.

This category also anticipates future developments: As adversaries employ AI, psychological techniques may target not only human operators but also training data and decision algorithms of adversary AI systems.

6.4 Taxonomy Labels: Goal and Placement

While our core taxonomy distinguishes techniques based on their operational mechanics, a purely technical classification is insufficient for operational planning. To solve the problem of fragmented terminology and to bridge the gap between academic research and industrial practice, we overlay our core structure with two standardized labeling dimensions: *Deception Goal* and *Deception Placement*. We deliberately selected the MITRE frameworks [16, 17] for three critical advantages: (1) leveraging existing practitioner familiarity rather than introducing new vocabulary, (2) enabling direct integration with existing operational workflows and tooling, and (3) benefiting from ongoing framework maintenance as MITRE evolves to address new threats. Defenders already using ATT&CK matrices can directly map our taxonomy into their existing processes, while security operations centers can incorporate deception techniques using the same vocabulary they employ for threat modeling and detection.

Deception Goal. We adopt the MITRE Engage matrix [17] to resolve the ambiguity of ad-hoc goal definitions in existing taxonomies. Rather than vague terms like *distraction* or *confusion*, MITRE Engage provides precise operations with defined success criteria such as *Expose* (reveal attacker presence, capabilities, or intent), *Affect* (Degrade attacker operations or decision-making), and *Elicit* (Induce specific attacker behaviors for observation). Techniques receive multi-label goal annotations, acknowledging that they often serve multiple objectives simultaneously. For example, a honeypot might support *Expose* (revealing presence upon access) and *Elicit* (inducing behaviors through crafted

vulnerabilities). This avoids problematic assumptions that single goals define primary classification, while enabling practitioners to select techniques aligned with specific defensive objectives.

Deception Placement. We utilize the MITRE ATT&CK matrix [16] to provide precise deployment vocabulary, replacing generic terms like *host-based* or *network-based* with detailed attack-stage and tactic-specific placement. MITRE ATT&CK maps the placement of defenses directly to the attack surface (e.g., Enterprise, Mobile, ICS). For example, honeytokens map directly to the Credential Access tactic. This alignment enables a clearer visualization of defensive coverage, helping practitioners identify which layers of their technology stack are protected by deceptive assets and where coverage gaps remain.

To support practical application, we have mapped these Goal and Placement labels to every specific technique within our taxonomy tree. As this detailed integration exceeds the spatial constraints of a static figure, the complete interactive mapping is available for consultation online (see Sec. 1 and Appendix 8).

7 CONCLUSION

This work presents the first systematic meta-survey of the cyber-deception literature, analyzing 46 survey papers to address the field’s growing fragmentation and provide a synthesis of its evolution, taxonomic approaches, and persistent research challenges. Our analysis reveals deep but narrow expertise across three research families: general cyber-deception, Moving Target Defense (MTD), and honeypots. The MTD literature exhibits strong taxonomic consensus yet remains fragmented across application domains, the honeypot literature continues to face long-standing evaluation challenges, and critical techniques such as tarpits, honey encryption, honey patches, and bias exploitation strategies remain largely absent from survey coverage.

To address these gaps, we introduced a unified taxonomy that integrates disparate classification schemes from across the literature. This taxonomy is grounded in the MITRE Engage and MITRE ATT&CK frameworks and is not merely a re-labeling of existing schemes, but is novel because it consolidates the fragmented taxonomies developed for general cyber-deception, MTD, and honeypots into a single cross-domain framework; it treats psychological and cognitively oriented deception—together with underrepresented techniques such as bias exploitation, tarpits, honey encryption, honey patches, and honeytokens—as first-class elements of the design space rather than peripheral add-ons; and it explicitly anchors deception goals and deployment locations in the MITRE Engage and MITRE ATT&CK matrices, so that they map cleanly onto vocabularies already used by practitioners for threat modeling, hunting, and incident response. Taken together, this taxonomy provides a common vocabulary that can bridge research silos and systematically organize both well-studied and under-explored deception techniques within a single coherent framework.

Our most significant cross-cutting finding is the lack of standardized evaluation frameworks and quantitative metrics. Despite repeated calls in the literature, the community still lacks rigorous methods to measure deception effectiveness and cost-benefit tradeoffs, and most work is validated only in small-scale or simulated settings. This limits comparability across techniques and weakens the empirical basis for deployment decisions.

The surveyed works converge on future directions such as AI- and ML-driven adaptive deception, hybrid combinations of complementary mechanisms, and tighter integration with existing security ecosystems. Our analysis highlights a central paradox: without resolving the evaluation gap, it remains difficult to determine whether such advances provide real security gains or simply add complexity. By synthesizing the survey landscape, identifying coverage and methodological gaps, and establishing a unified taxonomic framework, this meta-survey provides a consolidated map of the cyber-deception field and a foundation for more rigorous, measurable progress.

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8 APPENDIX

This appendix provides supplementary material to support the transparency and reproducibility of our methodology. In particular, we document the exact bibliographic search query used to identify candidate surveys and the critical appraisal questions that guided the initial screening and assessment of their methodological quality.

8.1 Scopus Search Strategy

Our starting point for identifying candidate defensive-deception surveys was the Scopus bibliographic index, as described in Section 2. The search string was designed around three main groups of terms:

- *Security context*: terms such as *cyber* and *security* to ensure that retrieved papers are grounded in cybersecurity or closely related domains.
- *Defensive-deception terminology*: linguistic variants capturing the core technique families of interest, including *deception*, *deceptive*, the stem *honey* (to cover honeypots, honeytokens, honeynets, etc.), *MTD* and the explicit phrase *moving target defense/defence*, as well as related concepts such as *tarpit*, *decoy*, *canary*, *canaries*, and *breadcrumb*.
- *Survey-type publication labels*: descriptors such as *survey*, *taxonomy*, *review*, *state of knowledge*, *SoK*, and *classification*, which explicitly indicate that the work is intended as a secondary study rather than a primary research contribution.

The query is restricted to the TITLE-ABS-KEY fields so that only terms appearing in the title, abstract, or author keywords are considered, which balances recall and precision and avoids relying on full-text availability. We explicitly include both American and British spellings of *defense/defence* to reduce language-induced omissions. The `PUBYEAR < 2025` predicate enforces the temporal cut-off of our study and matches the scope discussed in Section 2.

Applying this query in Scopus yielded 258 records before screening, which were then filtered according to the eligibility criteria and language restrictions presented in the main text. The exact Scopus query that was used for our methodology (cf. Section 2) is provided in Listing 1. We include it verbatim to enable direct reuse: future meta-surveys can, for example, extend the time window by adjusting the `PUBYEAR` constraint or augment the search with additional domain-specific keywords while preserving comparability with our study.

```
TITLE-ABS-KEY(
  ( cyber OR security )
  AND ( defense OR defend OR defence )
  AND ( deception OR deceptive OR honey OR mtd
        OR ( moving AND target AND defense )
        OR ( moving AND target AND defence )
        OR tarpit OR decoy OR canary OR canaries OR breadcrumb )
  AND ( survey OR taxonomy OR review OR ( state AND of AND knowledge )
        OR sok OR classification )
)
AND PUBYEAR < 2025
```

Listing 1. Scopus query for eligible candidate studies.

8.2 Quality Appraisal Questions

To support a consistent and transparent assessment of methodological quality, we employed a lightweight critical appraisal protocol during the initial screening of candidate surveys (cf. Section 2). The protocol is based on the Critical Appraisal Skills Programme (CASP) checklist for systematic reviews and the PRISMA reporting guidelines, which we adapted to the context of secondary studies on defensive deception and related security topics.

The resulting questions focus on three broad aspects:

- (1) **Basic study design:** whether the paper is structured in a way that is appropriate for a systematic review or mapping study, and whether it clearly articulates its research questions.
- (2) **Methodological soundness:** whether the literature search, screening, selection, and data-extraction procedures are described in sufficient detail to be reproducible and to justify the completeness of the evidence base.
- (3) **Trustworthiness of the results:** whether the synthesis, discussion, and reported limitations support a credible and useful interpretation of the evidence.

The full set of appraisal questions is listed in the table below. They are grouped into three sections that mirror these aspects: Section A addresses the basic validity of the study design; Section B focuses on the soundness and transparency of the review methodology; and Section C examines whether the reported results and limitations can be considered trustworthy and actionable.

ID	Critical Appraisal Questions for Initial Screening
Section A: Is the basic study design valid for a systematic review?	
A1	Did the systematic review address a clearly formulated research question?
A2	Did the researchers search for appropriate study design(s) to answer the research question?
Section B: Is the systematic review methodologically sound?	
B3	Were all the relevant primary research studies likely to have been included in the systematic review?
B3a	Searching for primary research studies.
B3b	Screening primary research studies from the search.
B3c	Selecting primary research studies to include in the systematic review.
B3d	Summarizing the search and its outputs.
B4	Did the researchers extract and present information from the individual primary research studies appropriately and transparently?
B4a	Extraction of data.
B4b	Presentation of data.
Section C: Are the results of the systematic review trustworthy?	
C5	Did the researchers analyse the pooled results of the individual primary research studies appropriately?
C5a	Subgroup analysis.
C6	Did the researchers report any limitations of the systematic review, and do these limitations cover all issues identified during critical appraisal?
C7	Would the benefits of acting upon the results outweigh any potential disadvantages, harms, or additional resource demands?

Table 8. Questions derived from CASP checklist for systematic review [20] and PRISMA checklist [64]

In our study, these questions were used as a qualitative checklist rather than as a numerical scoring rubric. During initial screening, each candidate survey was evaluated against the items in Table 8 to identify severe methodological weaknesses, such as the absence of an explicit search strategy, unclear inclusion and exclusion criteria, or a lack of discussion of limitations. These appraisals informed the exclusion of papers that were either mislabeled as surveys/taxonomies/SoKs or lacked any substantive treatment of defensive deception, and they provided structured support for the multi-reviewer conflict-resolution process described in Section 2.

Beyond informing inclusion and exclusion decisions, the checklist also served as an interpretive lens in our qualitative synthesis of the final set of surveys. In particular, it helped us to systematically characterize differences in methodological rigor across studies and to highlight recurring weaknesses in existing survey practice, as discussed in the main text. We make the checklist explicit here so that it can be reused or adapted by future meta-surveys in cyber-deception and related security domains.

8.3 Interactive Cyber Deception Matrix

To make our unified taxonomy practically usable, we provide an interactive web-based matrix² that we maintain as a living resource. The matrix instantiates the taxonomy presented in the main text (Sec. 6) and augments each concrete technique with metadata about its goals and deployment contexts.

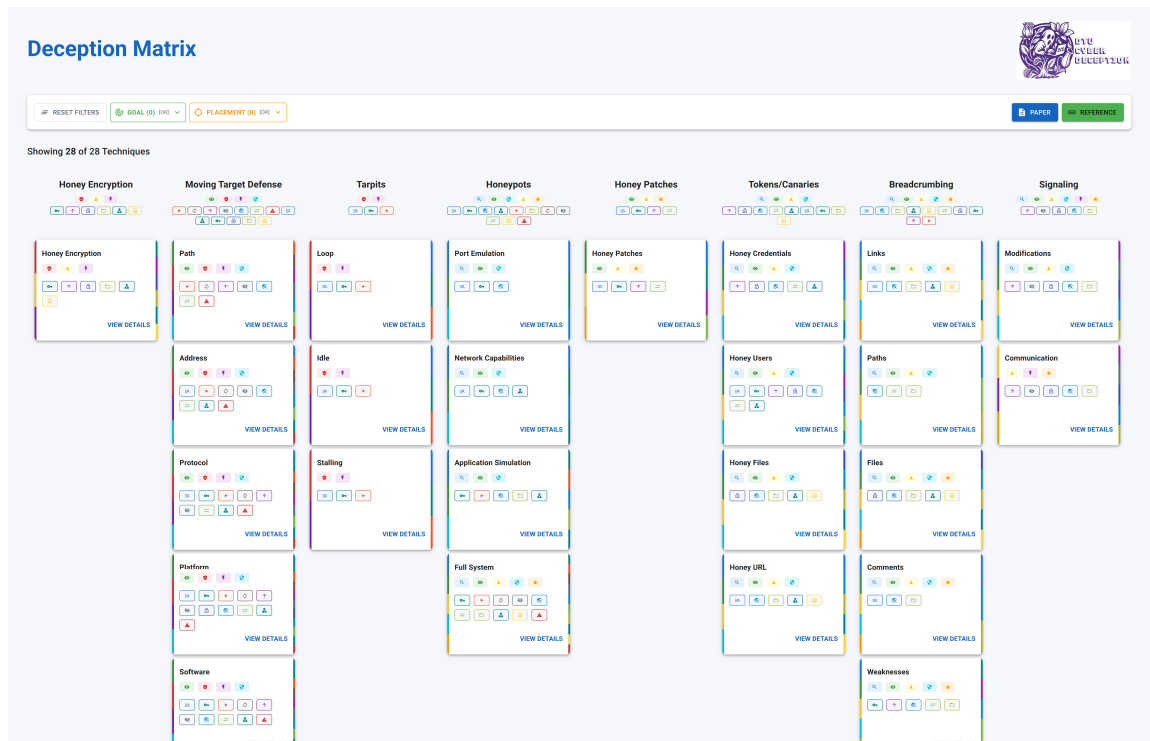


Fig. 3. Cyber deception taxonomy matrix: overview of the interactive web interface.

²<https://deception.compute.dtu.dk/matrix>

Each column in the matrix corresponds to a deception approach (for example, honeypots, honeytokens, tarpits, honey patches, or signaling), while the cards to the specific techniques. Each card summarizes: (i) the technique’s MITRE Engage goal labels, (ii) MITRE ATT&CK tactic(s), (iii) detailed description and (iv) the related resources. The interface supports filtering and search across goals and placements metadata, allowing users to restrict the view to techniques that, for instance, support a particular goal (such as *Expose* or *Affect*), are mapped to a specific ATT&CK tactic, or have publicly available implementations.

We include two screenshots to give a visual impression of the resource; however, we encourage readers to explore the online matrix directly. Figure 3 shows the default overview with multiple deception families visible, while Fig. 4 zooms in on the Tarpits and Stalling branch to illustrate how fine-grained variants are encoded and linked to the underlying literature and tools.

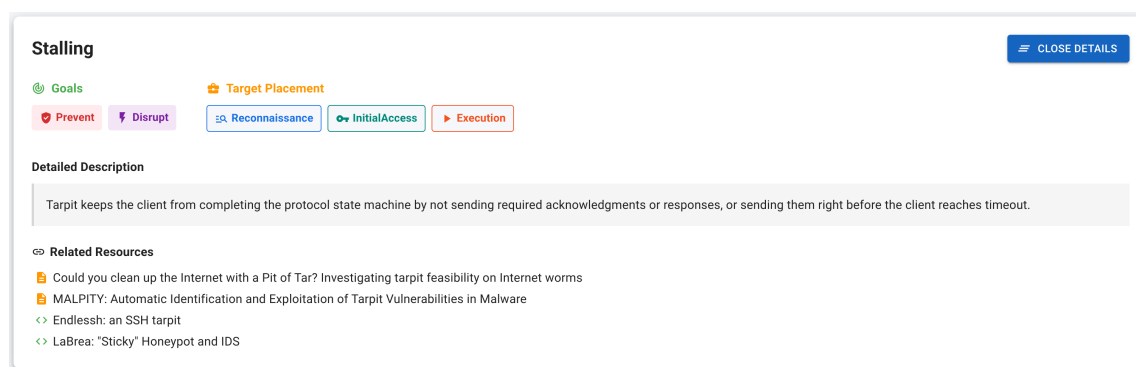


Fig. 4. Cyber deception taxonomy matrix: detailed view of tarpits and stalling techniques.